

TECHNICAL PROPOSAL AND SOFTWARE REQUIREMENTS AND DESIGN DOCUMENT

# Hadia Gifting Registry

Gift registry, item reservation and M-Pesa cash gifting platform with automated escrow and payouts.

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PREPARED FOR

Hadia  
Attn: Tracey, Founder

PREPARED BY

Codevertex Africa Limited  
[www.codevertexafrica.com](http://www.codevertexafrica.com)

DATE

6 October 2026

DELIVERY PERIOD

12 weeks

INVESTMENT

KES 240,000 build  
KES 8,000 per month

CONTACT

[info@codevertexafrica.com](mailto:info@codevertexafrica.com)

# Document control

| Version | Date           | Author                    | Description              |
|---------|----------------|---------------------------|--------------------------|
| 1.0     | 6 October 2026 | Codevertex Africa Limited | Issued for client review |

| Role           | Party      | Responsibility                                                   |
|----------------|------------|------------------------------------------------------------------|
| Product owner  | Hadia      | Scope approval, business decisions, legal sign-off, pilot events |
| Delivery lead  | Codevertex | Planning, scope control, reporting, risk management              |
| Technical lead | Codevertex | Architecture, code quality, security, releases                   |

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# Executive summary

Codevertex Africa Limited proposes to design, build and launch the Hadia Gifting Registry, a mobile-first platform on which registry owners list gifts for an occasion, share a single link, and receive item reservations and M-Pesa cash contributions from guests. Cash contributions are held in escrow and paid out to owners automatically, net of Hadia's commission, under rules configured by Hadia.

The platform is built on Codevertex's production services for identity, payments, accounting, escrow, approvals and messaging. The registry service, web application and administration console are developed specifically for Hadia.

## Key outcomes

- Registries published and shared within five minutes of sign-up, with no application download for owners or guests.
- Duplicate gifts eliminated through atomic item reservations visible to all guests in real time.
- M-Pesa contributions credited exactly once and reconciled hourly against the PayHero wallet.
- Payouts and commission deductions executed automatically by threshold and event date, with approval required only above Hadia's limit.
- Commission transferred to Hadia's business account daily without manual intervention.
- Compliance with the Data Protection Act, consumer protection, payments and accessibility requirements from launch.

## Engagement summary

|                                    |                                         |                                                                                            |                                                           |
|------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>12 weeks</b><br>Delivery period | <b>KES 240,000</b><br>Fixed build price | <b>KES 8,000</b><br>Monthly service fee:<br>treasury subscription and<br>dedicated support | <b>KES 12,000</b><br>Hosting per year, billed to<br>Hadia |
|------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------|

## SECTION 1

# Introduction

## 1.1 Purpose

This document specifies the requirements, design, delivery plan and commercial terms for the Hadia Gifting Registry. It is the basis for development, testing and acceptance.

## 1.2 Product scope

| Area              | Included                                                                                                                         |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Public website    | Landing page, how it works, occasions, help centre, legal pages, cookie consent                                                  |
| Owner application | Mobile sign-up, registries, items, sharing, gift log, thank-you tracking, payout settings, premium upgrade, privacy settings     |
| Guest experience  | Registry page, reservations, M-Pesa contributions (Airtel Money and card where enabled), receipts, reservation cancellation      |
| Funds management  | Escrow per registry, automated payouts, commission deduction and sweep, contribution limits, refunds, reconciliation, accounting |
| Administration    | Registry and user management, approvals, exception queue, reports, fee and limit settings, moderation, data requests             |
| Messaging         | SMS and WhatsApp receipts and alerts, email, marketing opt-in                                                                    |
| Compliance        | Legal policy drafts, consent records, data export and deletion, WCAG 2.2 AA accessibility                                        |

**Excluded:** native iOS and Android applications, wedding planning and RSVP tools, gift delivery and retail, retailer affiliate integrations, formal legal opinions, marketing campaigns.

## 1.3 Definitions

| Term           | Definition                                                                    |
|----------------|-------------------------------------------------------------------------------|
| Registry       | An owner's list of items and cash funds for an event                          |
| Escrow account | The treasury record holding a registry's cash contributions                   |
| Payout         | Transfer of a registry balance, net of commission, to the owner               |
| Commission     | Hadia's percentage fee on cash contributions, deducted at payout              |
| Sweep          | Scheduled transfer of earned commission to Hadia's business account           |
| Premium        | One-time paid upgrade per registry                                            |
| STK push       | M-Pesa payment prompt sent to the payer's phone                               |
| PayHero Team   | Hadia's PayHero account with an isolated wallet                               |
| KYC, AML       | Know Your Customer; anti-money laundering                                     |
| ODPC, DPIA     | Office of the Data Protection Commissioner; data protection impact assessment |
| eTIMS          | KRA electronic Tax Invoice Management System                                  |

|          |                                                   |
|----------|---------------------------------------------------|
| PWA      | Progressive web application                       |
| UAT      | User acceptance testing                           |
| RPO, RTO | Recovery point objective; recovery time objective |

## 1.4 References

- Hadia landing page concept and Gift Wishlist and Cash-Gifting Platform build report, August 2026
- Codevertex platform documentation: treasury escrow, PayHero integration, settlement flows, authentication, notifications
- Data Protection Act, 2019 and Data Protection (General) Regulations, 2021
- Web Content Accessibility Guidelines 2.2; OWASP Application Security Verification Standard 4

## SECTION 2

# Overall description

## 2.1 Product perspective

Hadia operates as a tenant of the Codevertex platform. Two new applications, hadia-ui and hadia-api, provide the registry product. Identity, payments, ledger, escrow, approvals and messaging are provided by existing platform services, described in [Solution architecture](#).

## 2.2 Product functions

- Registry creation and management for weddings, baby showers, housewarmings, graduations and birthdays
- Item listing by manual entry or product link, with reserve, fund or combined modes
- Guest reservations without duplicates, and cash contributions by M-Pesa
- Escrow of contributions with automated payouts and commission deduction
- Premium upgrade per registry: custom link, printable list, instant payouts
- Administration, reporting, reconciliation and compliance management

## 2.3 User classes

| Role                | Description                                       | Permissions                                                                                       | Authentication                 |
|---------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------|
| Guest               | Family, friends, diaspora relatives, colleagues   | View registry, reserve, contribute, leave a message, cancel own reservation                       | None; OTP when reserving       |
| Registry owner      | Couple, parent, graduate or host, aged 18 or over | Manage registries and items, share, view gifts, thank guests, set payout number, purchase premium | Mobile number and OTP          |
| Co-owner            | Invited by the owner                              | Edit items, thank guests                                                                          | Mobile number and OTP          |
| Hadia support       | Hadia staff                                       | User support, moderation, data requests                                                           | SSO with MFA                   |
| Hadia finance       | Hadia staff                                       | Approvals, ledger, reports, fees and limits                                                       | SSO with MFA; OTP per approval |
| Hadia administrator | Founder and delegates                             | All permissions, staff roles, settings                                                            | SSO with MFA                   |

## 2.4 Operating environment

- Guests and owners: any modern mobile or desktop browser; installable as a PWA
- Hosting: Codevertex Kubernetes platform with PostgreSQL, Redis, NATS JetStream and object storage, behind Cloudflare
- Payments: PayHero (M-Pesa, Airtel Money, card); messaging: Africa's Talking (SMS), Meta WhatsApp Cloud API, SMTP email

## 2.5 Constraints

- Funds are collected, held and disbursed only through a licensed payment service provider.
- The registry service does not move money; all payments are executed by the treasury service.

- Retailer product data is inconsistent; manual item entry is the primary method.
- Payment onboarding depends on Hadia's company registration and PayHero KYC.

## 2.6 Assumptions and dependencies

- Hadia is a registered Kenyan company with a KRA PIN and business bank account or paybill by week 2.
- Hadia completes PayHero KYC to tier 3 and provides requested documents within 3 working days.
- Hadia appoints a product owner with authority to decide within 2 working days.
- Hadia provides brand assets, approves copy and supplies consented testimonials.
- Hadia engages a Kenyan advocate to approve the legal pack and advise on payment regulation.
- Hadia recruits 3 to 5 pilot events for week 11.
- The treasury platform subscription, including escrow and M-Pesa features, and dedicated support are covered by the monthly service fee.
- Hadia bears hosting, domain, SMS, WhatsApp and payment provider charges as set out in the commercial terms.
- Hadia purchases and renews the domain name annually, or engages Codevertex to do so under the managed domain service.

## SECTION 3

# Functional requirements

Priority: **M** must have, **S** should have, **C** could have.

| ID                           | Requirement                                                                                                                                                                    | Pri. |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| ACCOUNTS AND ONBOARDING      |                                                                                                                                                                                |      |
| FR-01                        | Owners sign up and sign in with a mobile number and a 6-digit one-time password sent by SMS, with WhatsApp as fallback. Codes expire after 5 minutes. Sending is rate limited. | M    |
| FR-02                        | Owners confirm they are 18 or over and accept the terms and privacy notice. Version and timestamp are recorded.                                                                | M    |
| FR-03                        | Owners verify the payout M-Pesa number by one-time password. The registered M-Pesa name on the first payout is matched against the owner's name; mismatches are flagged.       | M    |
| FR-04                        | Owners invite co-owners by mobile number. Co-owners cannot change payout details.                                                                                              | S    |
| REGISTRIES AND ITEMS         |                                                                                                                                                                                |      |
| FR-05                        | Create a registry with type, title, date, optional location, cover photo and description. An owner may hold multiple registries.                                               | M    |
| FR-06                        | Add items manually: title, description, price, quantity, photo, store name and mode (reserve, fund or both).                                                                   | M    |
| FR-07                        | Add items from a product link. Public page metadata prefills the form for owner confirmation.                                                                                  | S    |
| FR-08                        | Create cash funds with an optional target.                                                                                                                                     | M    |
| FR-09                        | Reorder, edit, hide and archive items. Items with gifts can be archived but not deleted.                                                                                       | M    |
| FR-10                        | Registry visibility: public, unlisted (default) or passcode protected.                                                                                                         | S    |
| SHARING AND GUEST EXPERIENCE |                                                                                                                                                                                |      |
| FR-11                        | Each registry has a short link with a preview image, title and description for WhatsApp, Facebook and SMS.                                                                     | M    |
| FR-12                        | Guests view items, reservation status, fund progress and the event description without an account.                                                                             | M    |
| FR-13                        | Reservations are atomic; the last unit of an item cannot be reserved twice.                                                                                                    | M    |
| FR-14                        | Guests cancel their reservation from a link sent to them.                                                                                                                      | M    |
| FR-15                        | Owners control whether reserving guests' names are visible to other guests. Hidden by default.                                                                                 | S    |
| CONTRIBUTIONS AND PAYOUTS    |                                                                                                                                                                                |      |
| FR-16                        | Guests contribute toward an item or fund by M-Pesa STK push, within configured limits. Airtel Money and card are available where enabled.                                      | M    |
| FR-17                        | The registry receives the full gift. PayHero's 2.9% processing fee is shown before payment and added to the guest's total by default; Hadia can choose to absorb it instead.   | M    |
| FR-18                        | Where the STK push cannot be delivered, guests are offered a paybill and account number.                                                                                       | S    |
| FR-19                        | Guests receive an SMS or WhatsApp receipt. Owners receive a gift notification.                                                                                                 | M    |
| FR-20                        | Payouts run automatically when the balance reaches the threshold and after the event date. Premium registries pay out on each contribution.                                    | M    |

|                           |                                                                                                                                                                                                   |          |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>FR-21</b>              | Commission is deducted automatically at payout and itemised on every payout statement.                                                                                                            | <b>M</b> |
| <b>FR-22</b>              | Earned commission is transferred automatically to Hadia's business account on a schedule, above a minimum amount.                                                                                 | <b>M</b> |
| <b>FR-23</b>              | Payouts at or above the approval limit require finance approval with OTP. Payouts below the limit run without intervention.                                                                       | <b>M</b> |
| <b>FR-24</b>              | Cancelling a registry refunds each contribution to the paying number. Contributions without a number are listed for manual refund.                                                                | <b>M</b> |
| <b>FR-25</b>              | Owners download a statement of contributions and payouts in PDF or spreadsheet format.                                                                                                            | <b>S</b> |
| <b>OWNER TOOLS</b>        |                                                                                                                                                                                                   |          |
| <b>FR-26</b>              | Gift log with guest name, amount, item and message; thank-you status; WhatsApp thank-you message.                                                                                                 | <b>M</b> |
| <b>FR-27</b>              | Premium upgrade per registry: custom link, printable list, instant payouts.                                                                                                                       | <b>M</b> |
| <b>FR-28</b>              | Reminders to guest and owner 14 days after an unpurchased reservation. Owner may release the item.                                                                                                | <b>C</b> |
| <b>ADMINISTRATION</b>     |                                                                                                                                                                                                   |          |
| <b>FR-29</b>              | Search registries, owners and contributions. View a registry as a guest.                                                                                                                          | <b>M</b> |
| <b>FR-30</b>              | Configure commission, premium price, contribution limits, payout thresholds, approval limit and sweep schedule.                                                                                   | <b>M</b> |
| <b>FR-31</b>              | Exception queue: failed payouts, name mismatches, manual refunds, reconciliation variances, reported registries.                                                                                  | <b>M</b> |
| <b>FR-32</b>              | Content moderation with recorded reasons and owner notification.                                                                                                                                  | <b>M</b> |
| <b>FR-33</b>              | Reports: contributions and value by period, active registries, commission, premium sales, conversion, payout times.                                                                               | <b>M</b> |
| <b>FR-34</b>              | Data subject requests (access, correction, deletion) with deadline tracking.                                                                                                                      | <b>M</b> |
| <b>FINANCIAL CONTROLS</b> |                                                                                                                                                                                                   |          |
| <b>FR-35</b>              | Card contributions are held for 14 days before becoming eligible for payout. Chargebacks and reversals are debited from the registry, then the commission reserve, then recovered from the owner. | <b>M</b> |
| <b>FR-36</b>              | Scheduled promotional commission rates, including 0%, applied to registries created within the promotion window.                                                                                  | <b>S</b> |
| <b>FR-37</b>              | Hadia can freeze and unfreeze payouts per registry or platform-wide, and trigger manual payouts.                                                                                                  | <b>M</b> |
| <b>FR-38</b>              | Owners raise payout disputes from the dashboard; disputes enter the exception queue.                                                                                                              | <b>S</b> |

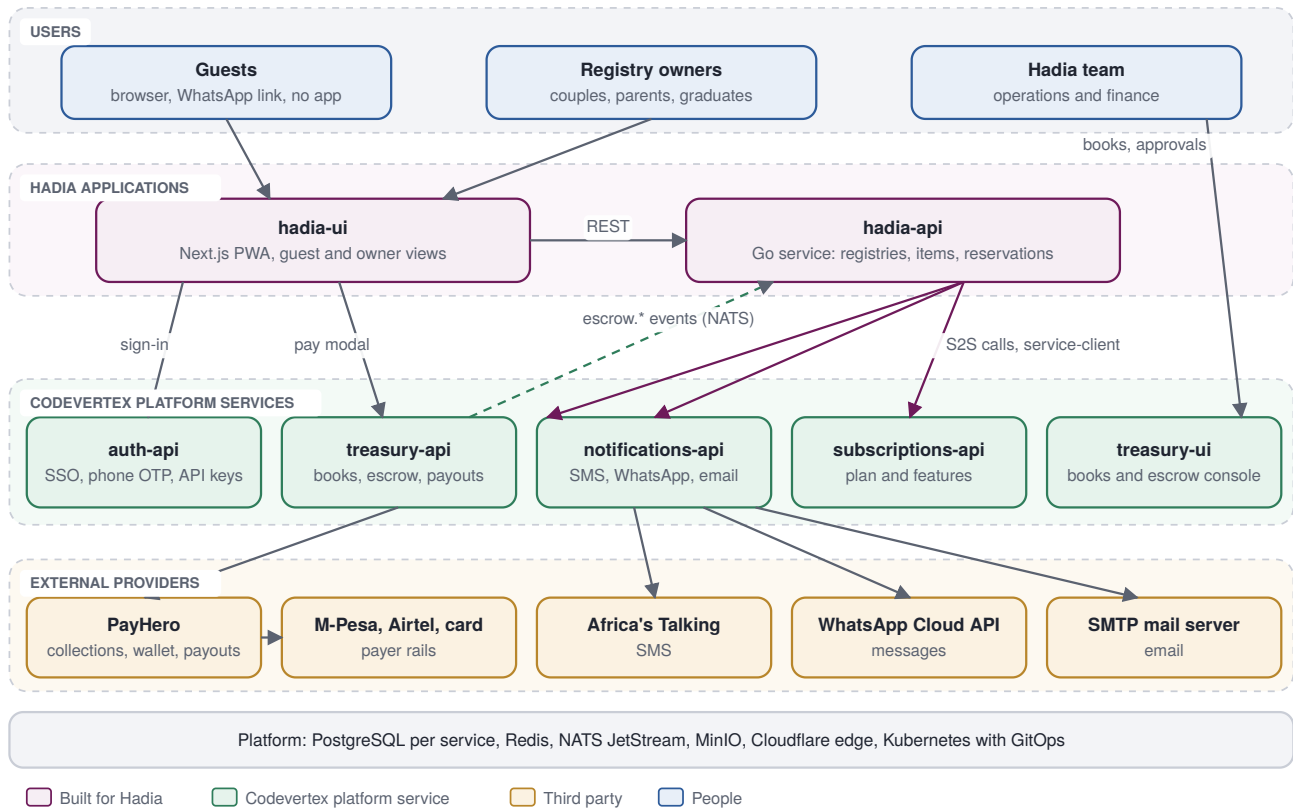
## SECTION 4

# Non-functional requirements

| ID            | Attribute                  | Requirement                                                                                                                                                                 |
|---------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NFR-01</b> | <b>Mobile performance</b>  | Guest page initial transfer under 150 KB excluding photos. Largest Contentful Paint under 2.5 seconds on a mid-range Android device over 3G. Photos served as resized WebP. |
| <b>NFR-02</b> | <b>Response time</b>       | 95th percentile under 300 ms for reads and 600 ms for writes, excluding payment provider latency.                                                                           |
| <b>NFR-03</b> | <b>Capacity</b>            | 10,000 active registries and 2,000 contributions per hour at peak, scalable tenfold without redesign. Keyset pagination on all lists; reports served from daily aggregates. |
| <b>NFR-04</b> | <b>Availability</b>        | 99.5% monthly for guest pages and checkout. Minimum two replicas per service; zero-downtime deployments.                                                                    |
| <b>NFR-05</b> | <b>Financial integrity</b> | Each contribution credited once, each payout executed at most once, balanced ledger, hourly reconciliation with the PayHero wallet.                                         |
| <b>NFR-06</b> | <b>Security</b>            | OWASP ASVS Level 2, TLS throughout, encrypted secrets, MFA for staff, audit trail on all financial and configuration changes.                                               |
| <b>NFR-07</b> | <b>Privacy</b>             | Data minimisation, consent records, data subject requests within statutory timelines, documented hosting location and transfer safeguards.                                  |
| <b>NFR-08</b> | <b>Accessibility</b>       | WCAG 2.2 Level AA.                                                                                                                                                          |
| <b>NFR-09</b> | <b>Localisation</b>        | English at launch. Text externalised for Kiswahili.                                                                                                                         |
| <b>NFR-10</b> | <b>Recovery</b>            | Daily encrypted offsite backups retained as 7 daily and 4 weekly copies; quarterly restore drills. RPO 24 hours, RTO 4 hours.                                               |
| <b>NFR-11</b> | <b>Observability</b>       | Structured logs, distributed tracing, alerts on failed payouts, reconciliation variances, error rates and latency.                                                          |
| <b>NFR-12</b> | <b>Maintainability</b>     | Codevertex platform stack and conventions, automated tests, documented APIs.                                                                                                |

## SECTION 5

# Solution architecture

**Figure 1.** Solution architecture

## 5.1 Components

| Component         | Status             | Function                                                                                                                                                                                              |
|-------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hadia-ui          | New                | Next.js progressive web app: website, guest pages, owner dashboard, administration console. Uses the Codevertex shared UI library for payments, sign-in, data tables, legal links and cookie consent. |
| hadia-api         | New                | Go service for registries, items, reservations, premium purchases, gift log, statistics and consent records. Does not move money.                                                                     |
| auth-api          | Existing, extended | Tenant, staff single sign-on with MFA, roles, terms acceptance, sessions. Extended with passwordless mobile sign-in for consumers.                                                                    |
| treasury-api      | Existing, extended | Ledger, PayHero gateway, checkout, escrow, approvals, payout dispatch, reconciliation, statements, eTIMS invoicing. Extended with scheduled commission sweep and per-registry contribution limits.    |
| treasury-ui       | Existing           | Finance console: ledger, escrow accounts, approvals, wallet, reports.                                                                                                                                 |
| notifications-api | Existing           | SMS (Africa's Talking), WhatsApp (Meta Cloud API), email (SMTP), push. Hadia message templates added.                                                                                                 |
| subscriptions-api | Existing           | Hadia's platform plan and feature entitlements.                                                                                                                                                       |

---

|                     |                 |                                                                                                                                                              |
|---------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shared<br>libraries | <b>Existing</b> | Authentication client, service client with retries and circuit breakers, event outbox, cache and job leases, pagination, HTTP middleware, shared UI library. |
|---------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## 5.2 Architecture principles

- **Single data owner.** hadia-api references treasury records by ID. Balances are read from treasury.
- **Single payment path.** Only treasury-api communicates with PayHero.
- **Event driven.** Treasury publishes escrow events through a transactional outbox to NATS. hadia-api consumes them idempotently.
- **Replica safe.** Scheduled jobs run once per period across all replicas using shared leases.
- **Standard stack.** Go with Ent and Atlas migrations, Next.js, Helm and ArgoCD.

## SECTION 6

# Funds management and payouts

## 6.1 Custody

Hadia operates as a tenant on the Codevertex platform with its own PayHero Team account, registered in Hadia's name and verified to PayHero KYC tier 3. The Team wallet is isolated and used only by Hadia. All contributions are received into this wallet. Treasury maintains a separate sub-ledger for each registry, so the wallet balance is fully allocated between registries and Hadia's earned commission at all times.

Codevertex does not receive or hold Hadia funds or customer funds.

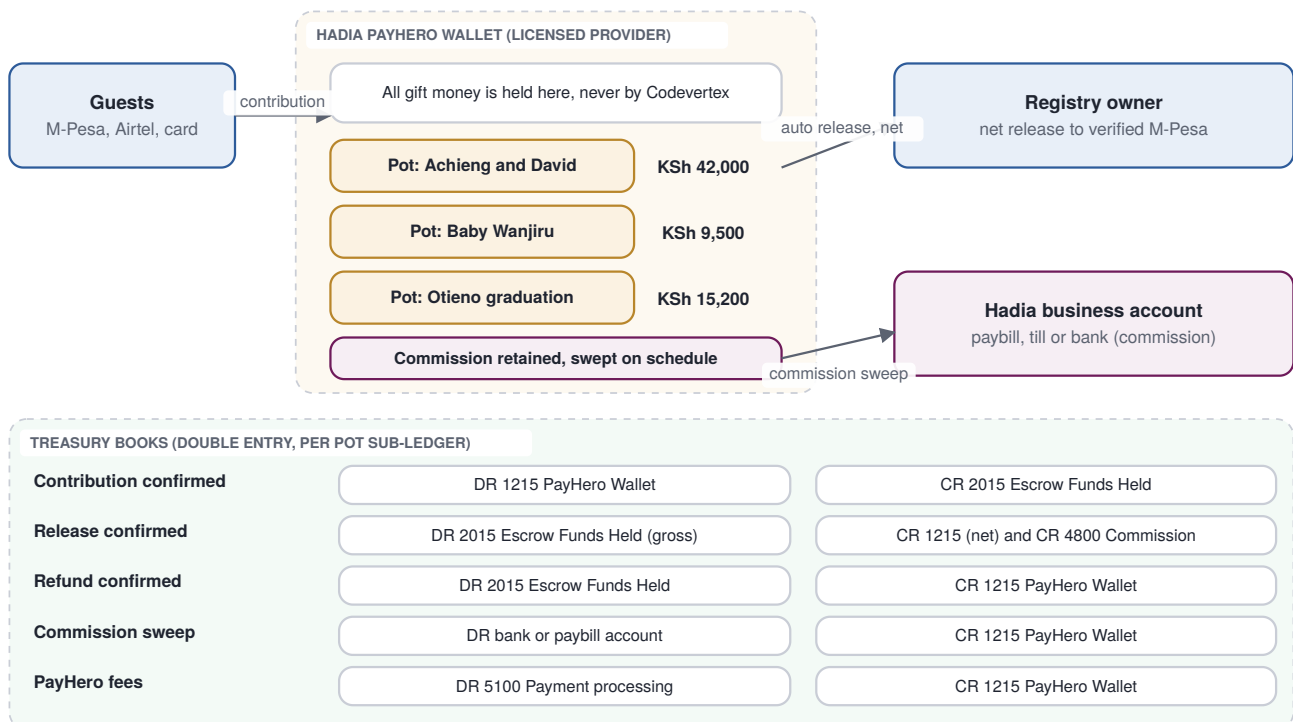


Figure 2. Funds flow and accounting entries

## 6.2 Registry escrow account

Each published registry has an escrow account in treasury. The account holds the registry title, the owner as beneficiary with a verified M-Pesa number, the target, payout rules and gift items. The registry ID is used as the external reference, which prevents duplicate accounts on retry.

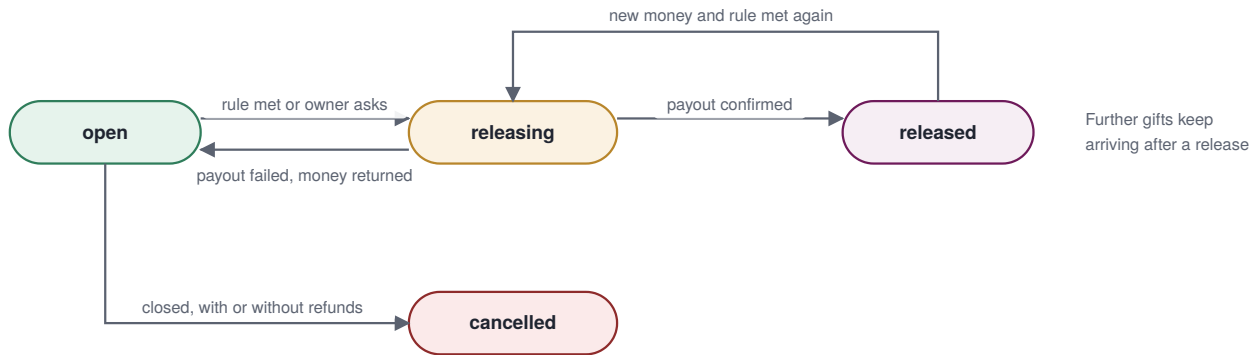


Figure 3. Escrow account lifecycle

### 6.3 Payout configuration

| Setting             | Level                         | Default                                             | Behaviour                                                                                        |
|---------------------|-------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Automatic payout    | Registry                      | On                                                  | Payouts run when a rule is met, checked every 5 minutes                                          |
| Payout threshold    | Registry                      | KES 10,000 standard; every contribution for premium | Full balance paid when the threshold is reached                                                  |
| Hold until          | Registry                      | Event date plus 2 days                              | Remaining balance paid after this date                                                           |
| Approval limit      | Hadia                         | KES 150,000 per payout                              | Payouts at or above the limit require finance approval with OTP, bound to amount and destination |
| Commission          | Hadia, with registry override | 4%                                                  | Deducted at payout and recorded as Hadia income                                                  |
| Contribution limits | Registry                      | KES 50 minimum, KES 150,000 maximum                 | Contributions outside the range are rejected before the STK push                                 |
| Maximum balance     | Platform                      | Agreed with Hadia                                   | Contributions exceeding the cap are declined                                                     |
| Maximum hold        | Platform                      | 180 days                                            | Funds cannot be held beyond this period                                                          |
| Commission sweep    | Hadia                         | Daily above KES 5,000, KES 2,000 reserve            | Earned commission transferred to Hadia's business account                                        |

## 6.4 Payout process

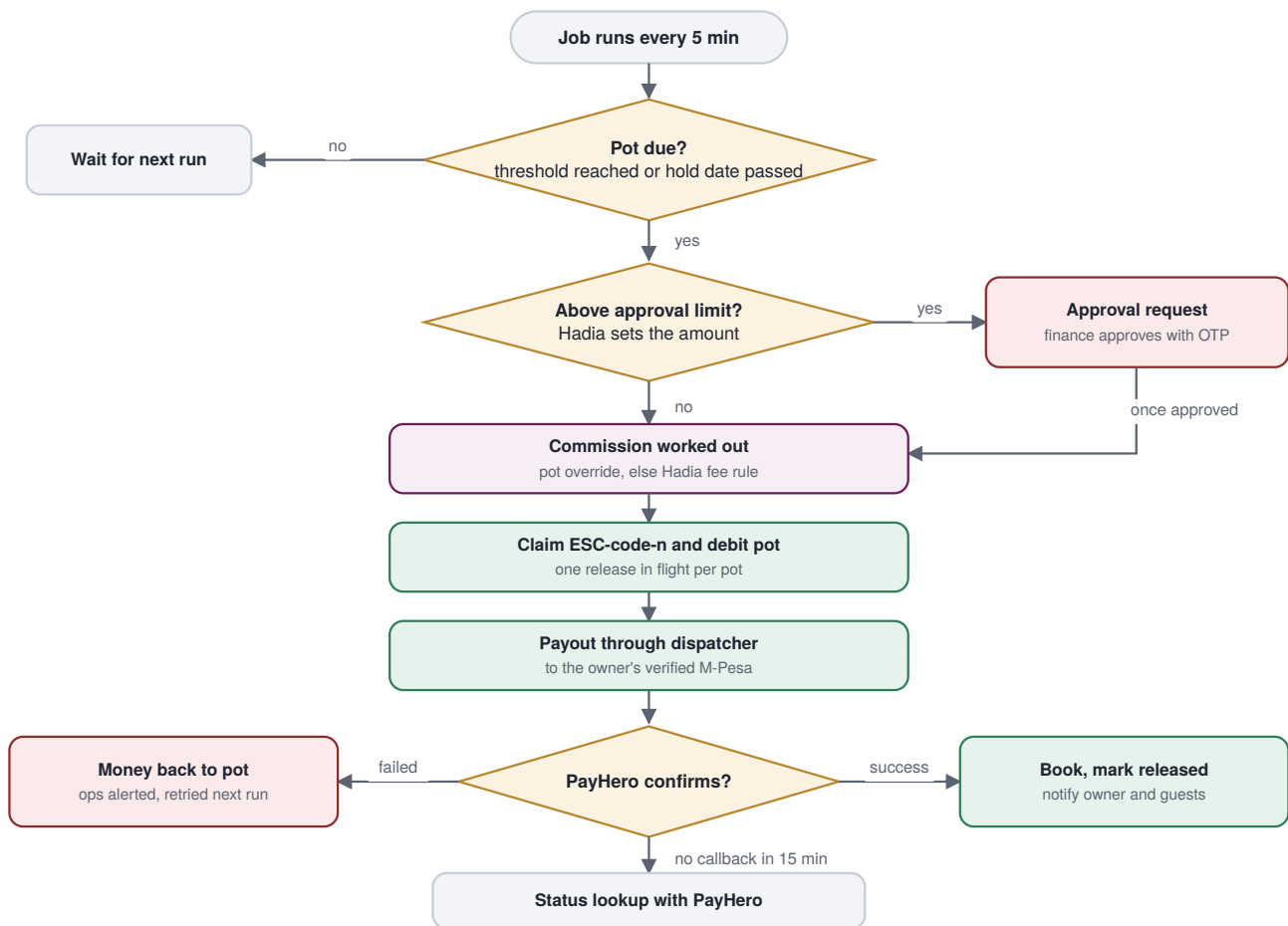


Figure 4. Automated payout process

Each payout attempt is registered under a unique reference before PayHero is called. Only one payout may be active per registry. A failed payout returns the funds to the registry balance. Payouts without a callback after 15 minutes are confirmed by status enquiry. All results are verified with PayHero before posting.

## 6.5 Commission

Commission is deducted from the gross payout and the net amount is paid to the owner. Commission is recorded in income account 4800 and retained in the Team wallet until swept.

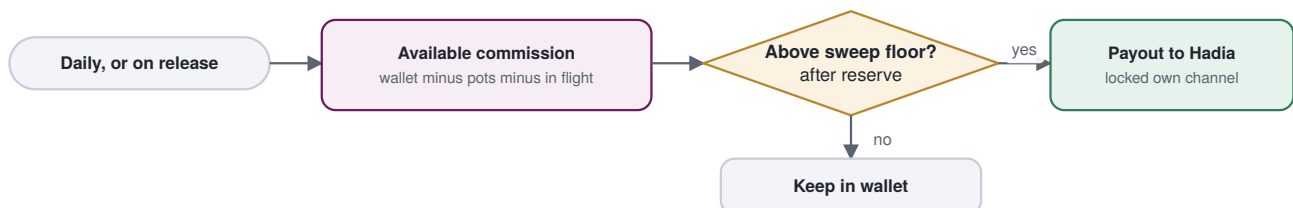


Figure 5. Commission sweep

1. The sweep runs daily, or after each payout if configured.
2. Available commission equals the wallet balance less registry balances, payouts in progress and the operating reserve.
3. Amounts above the sweep minimum are paid to Hadia's verified business channel. Changing the destination requires MFA and re-verification.
4. The transfer is posted from the PayHero wallet account to Hadia's bank or paybill account. The sweep is suspended if reconciliation reports a shortfall.

Manual withdrawal of commission remains available as an alternative.

## 6.6 Reconciliation

Treasury compares the total of registry balances and payouts in progress with the PayHero wallet balance every hour. Any shortfall is raised in the exception queue and suspends the commission sweep until resolved. Each registry statement is available in PDF, CSV and Excel.

## 6.7 Commission basis and configuration

| Question             | Position                                                                                                                                                                                                                     |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Basis of calculation | Commission is charged on the full gift. PayHero's processing fee never reduces the registry balance.                                                                                                                         |
| Worked example       | A guest gives KES 2,000 and pays KES 2,060, including PayHero's 2.9% processing fee. The registry is credited KES 2,000. At payout, 4% commission (KES 80) is deducted and the owner receives KES 1,920.                     |
| Processing fee       | Added to the guest's total by default and shown before payment. If Hadia chooses to absorb it, the guest pays KES 2,000 and Hadia covers the KES 58 fee from its commission.                                                 |
| Payout fee           | PayHero's payout fee is taken from the wallet and paid from Hadia's commission. The owner receives the full net amount.                                                                                                      |
| Changing the rate    | Hadia finance changes the commission rule (percentage, fixed amount, minimum and maximum fee) at any time from the administration console. Changes apply to payouts made after the change and are recorded in the audit log. |
| Per-registry rates   | A different rate, including 0%, can be set on individual registries, for example for partner planners or VIP clients.                                                                                                        |
| Promotional periods  | Hadia can schedule a promotion with a start date, end date and rate, including 0%. Registries created within the window keep the promotional rate for their lifetime.                                                        |

## 6.8 Payout controls

| Control                 | How it works                                                                                                                                                                                                                                                                              |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Freeze a payout         | Hadia can freeze any registry. Automatic payouts stop and the balance is held until the freeze is lifted. A platform-wide freeze switches every payout to manual approval.                                                                                                                |
| Manual payout           | Hadia staff, or the owner on request, can trigger an immediate payout of the available balance, subject to the approval limit.                                                                                                                                                            |
| Incorrect payout number | The payout number is verified by one-time password before the first payout, and the registered M-Pesa name is compared with the owner's name. A mismatch blocks the payout until reviewed. If funds still reach a wrong number, Hadia initiates a reversal through PayHero and Safaricom. |

|                |                                                                                                                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Payout dispute | The owner raises a dispute from the dashboard. The statement shows every contribution and payout with M-Pesa receipt numbers. Disputes enter the exception queue with a 2 working day response target. |
| Failed payout  | Funds return to the registry balance automatically, the owner is notified, and the payout is retried on the next cycle. Repeated failures move to the exception queue.                                 |

## 6.9 Reversals, fraud and chargebacks

| Event                              | Handling                                                                                                                                                                                                                                                                               | Loss allocation                                                        |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Payment reversed before payout     | The reversal is debited from the registry balance and the owner is notified.                                                                                                                                                                                                           | No loss                                                                |
| Payment reversed after payout      | Recorded as an amount owed by the owner and recovered from future contributions to any of the owner's registries. Unrecovered amounts are pursued under the owner terms.                                                                                                               | Hadia, pending recovery from the owner                                 |
| Card chargeback                    | Card contributions become eligible for payout only after a 14-day hold. A chargeback is debited from the registry balance first, then from Hadia's commission reserve, then recovered from the owner. Hadia responds to the dispute with the contribution evidence held in the ledger. | Hadia, pending recovery                                                |
| Fraudulent or unauthorised payment | Frozen on detection; refunded to the payer where confirmed. Monitoring flags unusual patterns such as many large gifts from few numbers.                                                                                                                                               | Hadia as operator, except where caused by a Codevertex software defect |

M-Pesa STK payments are authorised by the payer's PIN, so unauthorised M-Pesa contributions are rare. Card chargeback rights last up to 120 days under card scheme rules, which is why card-funded gifts carry the 14-day hold and a commission reserve.

## 6.10 Payment failure scenarios

| Scenario            | Guest experience                                                        | System behaviour                                                        |
|---------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|
| M-Pesa unavailable  | Prompt fails; guest is offered the paybill option, Airtel Money or card | Contribution held as pending; no duplicate charge on retry              |
| PayHero unavailable | Contributions temporarily unavailable; reservations continue            | Payouts queue and resume automatically; funds remain safe in the wallet |
| Payout fails        | Owner notified of the delay                                             | Funds returned to the registry; retried next cycle                      |
| Registry cancelled  | Guests notified that gifts are being returned                           | Each contribution refunded to the paying number                         |

## 6.11 International guests

| Question            | Position                                                                                                                                                                  |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Visa and Mastercard | Supported through PayHero's hosted card checkout, subject to the card rail being enabled on Hadia's PayHero account at onboarding. Card data never touches Hadia systems. |
| Countries           | Cards: guests worldwide, subject to issuer approval. Mobile money: the 14 countries listed below.                                                                         |
| Currencies          | Registries are held in Kenya shillings. Mobile money outside Kenya is charged in the local currency; cards are charged in KES.                                            |

|                              |                                                                                                                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Foreign exchange             | Cards: the guest's card issuer converts at its rate. Mobile money: treasury converts at a stored exchange rate refreshed every 6 hours and rounded up; the conversion is recorded on each payment. |
| Fees to international guests | Hadia charges guests nothing. The guest's bank may apply its own foreign transaction fee. PayHero's card processing charge is borne by Hadia.                                                      |

## 6.12 Supported mobile money countries

| Region          | Country                          | Charged in |
|-----------------|----------------------------------|------------|
| East Africa     | Kenya                            | KES        |
| East Africa     | Uganda                           | UGX        |
| East Africa     | Tanzania                         | TZS        |
| East Africa     | Rwanda                           | RWF        |
| East Africa     | Burundi                          | BIF        |
| East Africa     | Ethiopia                         | ETB        |
| Central Africa  | Democratic Republic of the Congo | CDF        |
| Southern Africa | Zambia                           | ZMW        |
| Southern Africa | Malawi                           | MWK        |
| Southern Africa | Mozambique                       | MZN        |
| Southern Africa | Botswana                         | BWP        |
| Southern Africa | South Africa                     | ZAR        |
| West Africa     | Nigeria                          | NGN        |
| West Africa     | Ghana                            | GHS        |

Kenya is live at launch through M-Pesa and Airtel Money. Each other country is enabled when PayHero activates the mobile money rail for that country on Hadia's account; networks offered (for example MTN, Airtel) follow PayHero's coverage in each country.

## SECTION 7

# Process workflows

Each workflow defines its trigger, steps, automation and exception handling. Steps marked **AUTO** run without user or staff action.

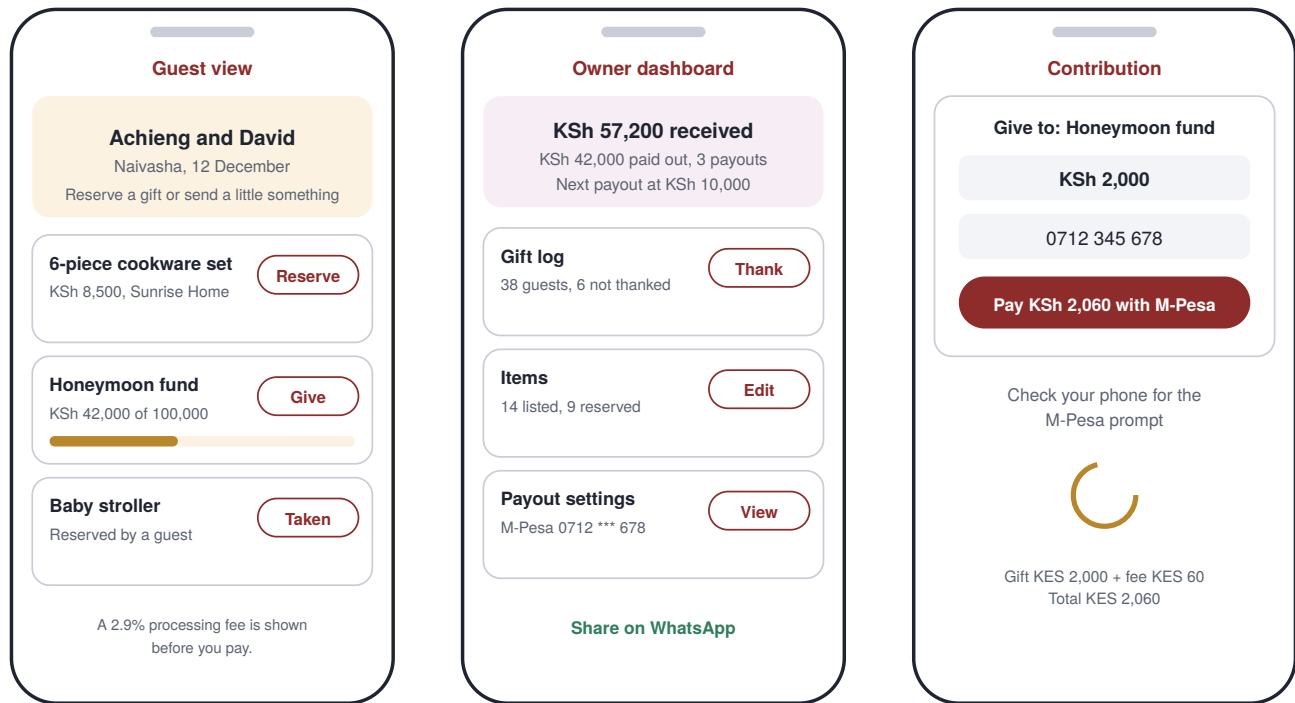


Figure 6. Key screens (low fidelity)

## 7.1 Hadia business onboarding



Figure 7. Hadia business onboarding

| Step | Action                                                                                                                   | Responsible    |
|------|--------------------------------------------------------------------------------------------------------------------------|----------------|
| 1    | Submit company documents: certificate of incorporation, CR12, KRA PIN, director identification, bank or paybill details. | Hadia          |
| 2    | Create the Hadia tenant and assign the platform plan with escrow and M-Pesa features.                                    | Codevertex     |
| 3    | Set up Hadia's PayHero Team, complete KYC to tier 3 and save the Team's API key in treasury.                             | Hadia, PayHero |
| 4    | Synchronise the paybill or till and map the wallet and channels to the chart of accounts.                                | Codevertex     |
| 5    | Accept the escrow terms in treasury.                                                                                     | Hadia          |
| 6    | Configure commission, payout policy, approval limit, contribution limits and sweep schedule.                             | Hadia finance  |

|   |                                                                                          |                      |
|---|------------------------------------------------------------------------------------------|----------------------|
| 7 | Execute live KES 1 tests for contribution, payout, refund and sweep, and reconcile each. | Codevertex,<br>Hadia |
|---|------------------------------------------------------------------------------------------|----------------------|

## 7.2 Registry owner onboarding

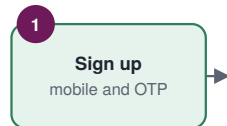


Figure 8. Registry owner onboarding

| Step | Action                               | System behaviour                                                                                                             |
|------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1    | Owner enters a mobile number         | One-time password sent by SMS, WhatsApp fallback. Codes hashed, 5-minute expiry, rate limited per number and IP. <b>AUTO</b> |
| 2    | Owner confirms age and accepts terms | Version and timestamp recorded                                                                                               |
| 3    | Owner creates the registry           | Type, title, date, location, cover photo, description                                                                        |
| 4    | Owner verifies the payout number     | One-time password sent to the payout number                                                                                  |
| 5    | Registry published                   | Escrow account opened in treasury with default rules <b>AUTO</b>                                                             |
| 6    | Owner shares the registry            | Short link and preview image generated <b>AUTO</b>                                                                           |

**Exception:** if the escrow account cannot be opened, the registry is saved as a draft and the request is retried in the background. Item entry continues uninterrupted.

## 7.3 Item listing

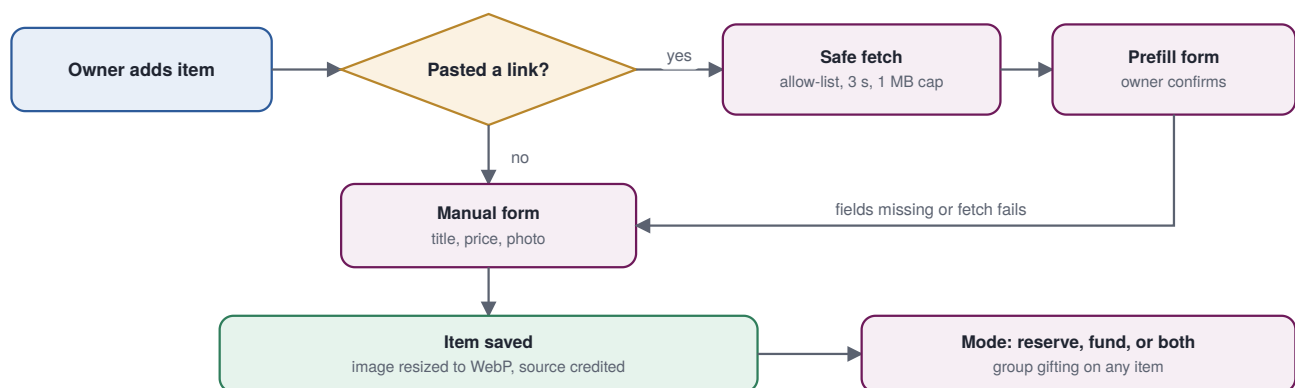


Figure 9. Item listing

1. The owner pastes a product link or selects manual entry.
2. For links, the server retrieves public page metadata under strict controls: public addresses only, 3-second timeout, 1 MB limit, no cookies.
3. Title, image and price are prefilled for owner confirmation. Where no data is found, the manual form is shown.
4. Photos are converted to WebP at 640 and 1280 pixels, stored in object storage and served through the CDN. The source store is credited.
5. The owner sets the item mode: reserve, fund, or both.

## 7.4 Sharing

1. Each registry receives a short link; premium registries may set a custom link.
2. A 1200 by 630 pixel preview image under 300 KB is generated for WhatsApp and social platforms. **AUTO**
3. Sharing options: WhatsApp, SMS, copy link and QR code.

## 7.5 Item reservation

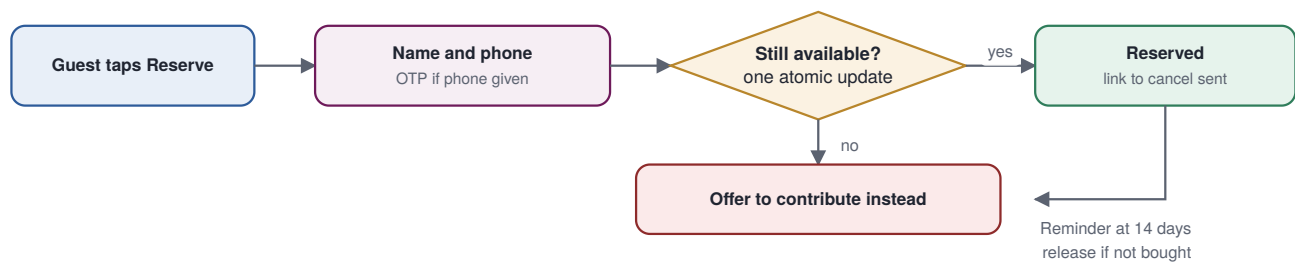
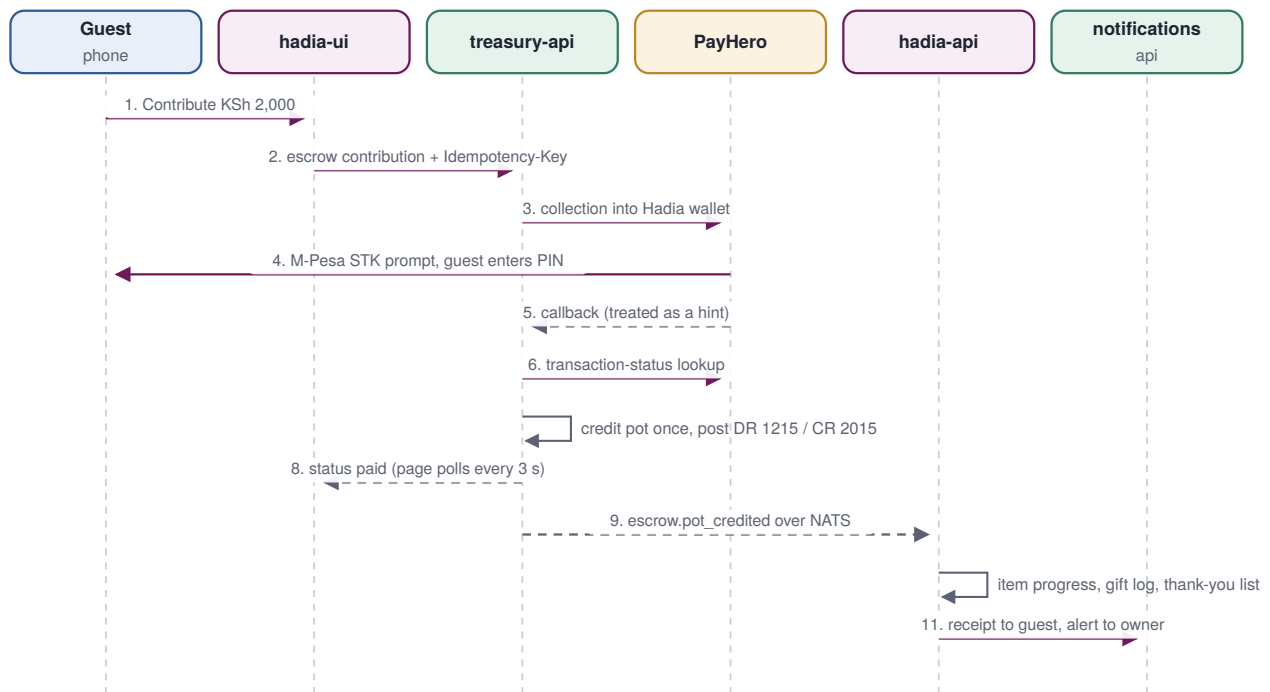


Figure 10. Item reservation

1. The guest selects Reserve and enters a name and, optionally, a mobile number.
2. The reservation is recorded in a single conditional update. Concurrent requests for the last unit result in one reservation; other guests are offered a contribution instead.
3. The guest receives a cancellation link. The item is shown as reserved to all guests.
4. After 14 days without purchase confirmation, reminders are sent and the owner may release the item. **AUTO**

## 7.6 Cash contribution



**Figure 11.** Cash contribution sequence

1. The guest selects an item or fund and enters an amount and mobile number. Limits are validated and the total, including any processing fee, is shown.
2. hadia-ui submits the contribution to treasury with an idempotency key.
3. Treasury initiates a PayHero collection into the Hadia wallet. The guest authorises the STK push.
4. On callback, treasury verifies the transaction with PayHero, credits the registry once and posts the ledger entry. **AUTO**
5. The checkout page confirms payment. Treasury publishes the credit event.
6. hadia-api updates item progress and the gift log and triggers the guest receipt and owner notification. **AUTO**

**Exceptions:** failed or cancelled prompts leave the contribution pending and the paybill option is offered. Repeated failures to one number (five in 24 hours) suspend further prompts to that number. Missing callbacks are resolved by status enquiry.

## 7.7 Registry management

1. The dashboard shows amounts received, paid out, pending and the next payout.
2. Owners may change the target, threshold (within Hadia's limits) and event date. The payout number changes only after verification of the new number and never during an active payout.
3. Contributions above an item's price remain in the registry balance and are paid out with it.
4. After the hold period the registry closes to new contributions. Owners may reopen it.

## 7.8 Registry lifecycle rules

| Question               | Rule                                                                                                                                                                                                                               |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purchase confirmation  | The reservation message contains a link to mark the gift as purchased, with an optional receipt photo. The owner sees "Reserved" or "Purchased" and can confirm receipt. Unconfirmed reservations trigger reminders after 14 days. |
| Reserved items view    | The owner dashboard filters items by Available, Reserved, Purchased and Funded, with the reserving guest's name and date.                                                                                                          |
| Manual release         | The owner can release any reservation. The guest is notified and the item becomes available again.                                                                                                                                 |
| Registry transfer      | Ownership can be transferred to another person. The new owner verifies their mobile number and payout number by one-time password. Transfers are blocked while a payout is in progress and are recorded in the audit log.          |
| Multiple beneficiaries | Each registry pays out to one verified M-Pesa number. Co-owners can manage the registry but payouts go to the designated beneficiary. Split payouts across several beneficiaries can be added in a later phase.                    |
| Event date change      | Allowed at any time, including after contributions begin. The payout hold date moves with the event date, within the 180-day maximum hold. Guests are not charged again and existing contributions are unaffected.                 |
| Registry expiry        | 30 days after the event date the registry closes to new contributions and reservations. Any remaining balance is paid out automatically. The registry stays visible to the owner in read-only mode.                                |
| Reopening              | A closed registry can be reopened by the owner within 90 days, or by Hadia at any time. Cancelled registries cannot be reopened.                                                                                                   |

## 7.9 Automated payout

Registries meeting a payout rule are paid every 5 minutes to the verified number, net of commission, with approval required only at or above Hadia's limit. The owner receives an SMS showing gross amount, commission and net amount. **AUTO** Full process in [Funds management and payouts](#).

## 7.10 Commission and sweep

Commission is deducted at payout, recorded as Hadia income and transferred daily to Hadia's business account above the sweep minimum. **AUTO** eTIMS invoices for Hadia's fees are issued through treasury's KRA integration.

## 7.11 Premium upgrade

1. The owner selects Upgrade and is shown the features and price.
2. hadia-api creates a payment intent in treasury and opens the shared payment modal.
3. On confirmation, treasury records the sale as Hadia revenue and hadia-api enables the custom link, printable list and instant payouts. **AUTO**
4. A receipt is issued. Refunds follow the refund policy.

## 7.12 Cancellation and refunds

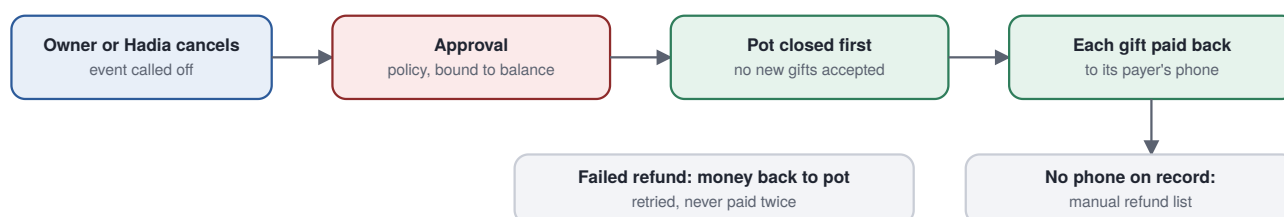


Figure 12. Cancellation and refunds

1. The owner requests cancellation, or Hadia cancels following a fraud or policy finding.
2. The refund is authorised under the approval policy, bound to the registry and its balance.
3. The registry is closed to new contributions. Each contribution is refunded to the paying number under its own reference. **AUTO**
4. Failed refunds return to the registry balance for retry. Repeated cancellation requests never duplicate a refund. Contributions without a number are listed for manual settlement.

## 7.13 Exceptions and reconciliation

1. Reconciliation runs hourly. **AUTO**
2. The exception queue lists failed payouts, name mismatches, manual refunds, reconciliation variances, reported registries and pending approvals.
3. Each entry records the facts, recommended action and resolver. Finance receives a daily summary.

## 7.14 Thank-you tracking

1. All contributions and reservations appear in the gift log with guest messages.
2. The owner sends a prepared WhatsApp thank-you and marks the guest as thanked.
3. A reminder listing outstanding thank-yous is sent one week after the event. Owners may disable it.

## 7.15 Data subject requests

1. An owner or guest submits a request for access, correction or deletion.
2. Identity is confirmed by one-time password and the statutory deadline is recorded.
3. Access requests produce a machine-readable export.
4. Deletion erases or anonymises personal data. Financial records required by law are retained for the statutory period under a reference, and the requester is informed.

## SECTION 8

# Data design

hadia-api owns the registry domain and references other services by ID: the owner's user ID in auth-api and the escrow account and payment intent IDs in treasury.

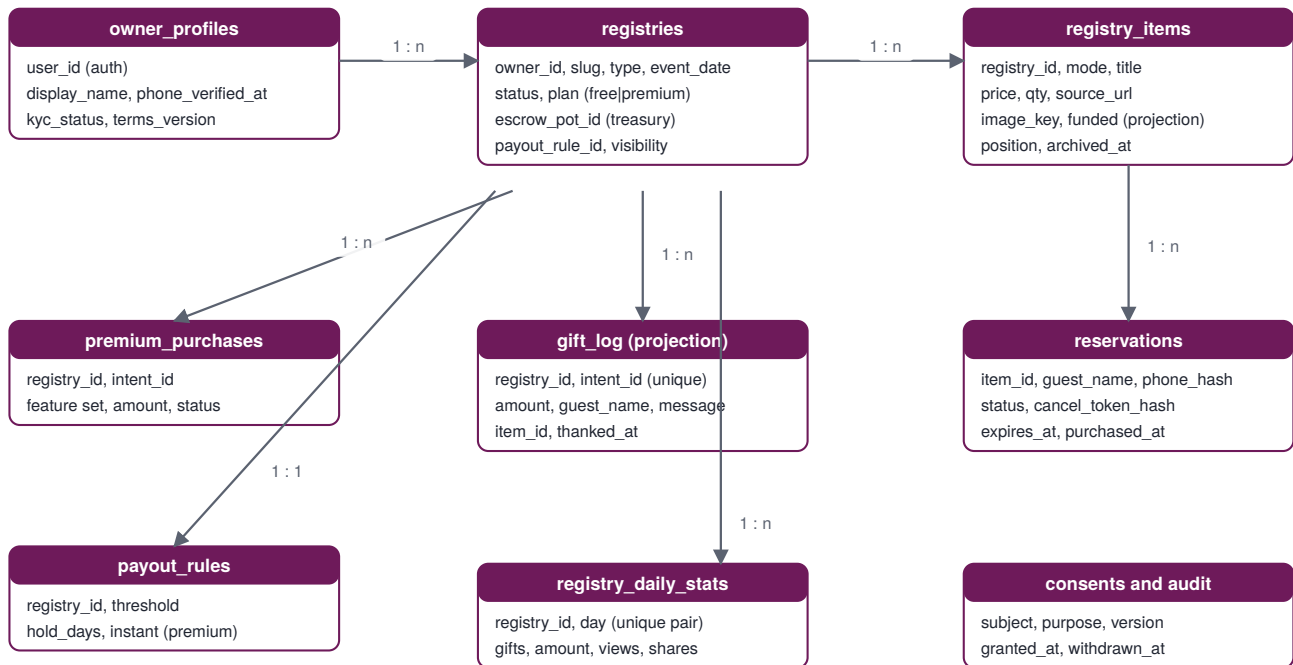


Figure 13. hadia-api data model

## 8.1 Scalability measures

| Measure                                                                                    | Purpose                                                   |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Tenant ID and query-specific indexes on every table, managed by versioned Atlas migrations | Consistent query performance and controlled schema change |
| Keyspace pagination on all lists                                                           | Constant response time regardless of page depth           |
| Unique constraint on payment intent ID in the gift log                                     | No duplicate records from redelivered events              |
| Counters updated in a single SQL statement                                                 | Correct totals under concurrency                          |
| Daily aggregate table for reporting                                                        | Reports independent of raw data volume                    |
| Guest numbers hashed for matching and encrypted where display is required                  | Data protection                                           |
| Images in object storage; keys only in the database                                        | Small, fast database                                      |

## SECTION 9

# Interfaces, events and notifications

## 9.1 hadia-api endpoints

| Area           | Endpoints                                                                                             | Access               |
|----------------|-------------------------------------------------------------------------------------------------------|----------------------|
| Registries     | GET, POST /registries; GET, PATCH /registries/{id}; POST /registries/{id}/publish, /close, /cancel    | Owner                |
| Items          | GET, POST /registries/{id}/items; PATCH, DELETE /items/{id}; POST /items/preview; POST /items/reorder | Owner                |
| Public         | GET /r/{code}; POST /r/{code}/items/{item}/reserve; POST /reservations/{token}/cancel                 | Public, rate limited |
| Gifts          | GET /registries/{id}/gifts; POST /gifts/{id}/thanked; GET /registries/{id}/statement                  | Owner                |
| Premium        | POST /registries/{id}/premium                                                                         | Owner                |
| Privacy        | POST /privacy/export, /privacy/delete; GET /privacy/requests                                          | Owner; guest by OTP  |
| Administration | Search, moderation, exceptions, settings, reports                                                     | Staff by role        |

## 9.2 Treasury endpoints

| Function                            | Endpoint                                                                       |
|-------------------------------------|--------------------------------------------------------------------------------|
| Escrow account create, read, update | POST, GET, PUT /{tenant}/escrow/pots                                           |
| Guest contribution                  | POST /api/v1/pay/{tenant}/escrow/{code}/contributions                          |
| Public account view                 | GET /api/v1/pay/{tenant}/escrow/{code}                                         |
| Payout; cancellation with refunds   | POST /{tenant}/escrow/pots/{pot}/release; /cancel?refund=true                  |
| Commission rule, readiness, summary | GET, PUT /{tenant}/escrow/fee-rule; GET /escrow/readiness; GET /escrow/summary |
| Statements                          | GET /{tenant}/escrow/pots/{pot}/statement; /statement/document                 |
| Premium sale                        | POST /{tenant}/payments/intents                                                |

## 9.3 Events

| Event                  | Publisher | hadia-api action                                                            |
|------------------------|-----------|-----------------------------------------------------------------------------|
| escrow.pot_created     | treasury  | Mark escrow ready; publish registry                                         |
| escrow.pot_credited    | treasury  | Update gift log, item funding and statistics; send receipt and notification |
| escrow.dispatched      | treasury  | Record payout; notify owner of gross, commission and net                    |
| escrow.dispatch_failed | treasury  | Raise exception; notify owner of retry                                      |
| escrow.cancelled       | treasury  | Close registry; notify guests of refunds                                    |

|                                                        |           |                                         |
|--------------------------------------------------------|-----------|-----------------------------------------|
| payment.succeeded                                      | treasury  | Activate premium features               |
| hadia.registry_published,<br>hadia.reservation_created | hadia-api | Consumed by notifications and analytics |

Consumers are idempotent on event ID. Events are retained in the outbox during broker outages.

## 9.4 Notifications

| Message                             | Recipient       | Channel                              |
|-------------------------------------|-----------------|--------------------------------------|
| One-time password                   | Owner, guest    | SMS; WhatsApp fallback               |
| Contribution receipt                | Guest           | SMS or WhatsApp                      |
| New contribution or reservation     | Owner           | WhatsApp or SMS; daily digest option |
| Payout completed or delayed         | Owner           | SMS                                  |
| Reservation and thank-you reminders | Guest, owner    | WhatsApp or SMS                      |
| Refund completed                    | Guest           | SMS                                  |
| Exception summary, approval request | Hadia finance   | Email, WhatsApp                      |
| Product updates                     | Opted-in owners | Email or WhatsApp with unsubscribe   |

## SECTION 10

# Reporting and analytics

| Report                                        | Audience         | Source                                              |
|-----------------------------------------------|------------------|-----------------------------------------------------|
| Registry dashboard                            | Owner            | Treasury balances and payouts; gift log             |
| Contributions and value by period             | Hadia            | Daily aggregates                                    |
| Active registries, new registries, conversion | Hadia            | Daily aggregates and page view counters             |
| Commission and premium revenue                | Hadia finance    | Treasury ledger, trial balance, income statement    |
| Funds held, in progress and available         | Hadia finance    | Treasury escrow summary                             |
| Payout performance                            | Hadia operations | Payout records: time to completion, failure reasons |
| Registry statement                            | Owner, finance   | Treasury statement in PDF, CSV, Excel               |

Analytical queries run on aggregate tables and may use the read replica. Dashboard results are cached for 60 seconds and invalidated by the related event.

## SECTION 11

# Security

| Threat                             | Controls                                                                                                                                                   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Account takeover                   | OTP rate limits by number and IP, hashed codes, short expiry, session revocation, MFA for staff, re-verification for payout changes                        |
| Payout redirection                 | Payout number verified by OTP, changes blocked during active payouts, approvals bound to amount and destination, name check on first payout, change alerts |
| Duplicate charges or payouts       | Idempotency keys, unique payout references, single settlement path, provider verification of callbacks                                                     |
| Fraudulent registries              | Moderation, guest reporting, registry creation limits, balance caps, first-payout hold for new owners                                                      |
| Money laundering                   | Contribution and balance limits, KYC, verified payout numbers, pattern monitoring, escalation to Hadia's compliance contact                                |
| Server-side request forgery        | Public address allow-list with post-resolution DNS checks, redirect validation, time and size limits                                                       |
| Injection and cross-site scripting | Parameterised queries, output encoding, Content Security Policy, sanitised input                                                                           |
| Automated abuse                    | Rate limiting, STK prompt guard, Cloudflare WAF                                                                                                            |
| Data exposure                      | TLS, encryption at rest, least-privilege roles, payout details excluded from public pages, staff access logging                                            |
| Card data                          | Provider-hosted card checkout; no card data on Hadia systems                                                                                               |

## 11.1 Production access

| Control               | Implementation                                                                                                                                            |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Who has access        | A named list of Codevertex platform engineers, provided to Hadia and updated within 5 working days of any change. No shared accounts.                     |
| Server access         | Key-based SSH only, restricted by host firewall (default deny; only SSH, HTTP and HTTPS open)                                                             |
| Changes to production | Only through version-controlled GitOps deployment (ArgoCD); manual changes are reverted automatically. Every change is attributable to a reviewed commit. |
| Secrets               | Encrypted as sealed secrets; separate database credentials per service; payment provider keys encrypted in the database                                   |
| Network isolation     | Kubernetes network policies on payment, authentication and notification services; pod security standards enforced                                         |
| Hadia staff           | Role-based access through single sign-on with MFA; finance approvals require a one-time password                                                          |
| Audit                 | Staff logins, approvals, settings changes and financial actions recorded in tamper-evident audit logs, available to Hadia                                 |

## 11.2 Codevertex access to Hadia data

Codevertex staff do not access Hadia's personal or financial records in normal operation. Access occurs only to resolve a support request raised by Hadia, to investigate an incident, or to perform a restore, and is limited to the records concerned. Each such access is logged and reported to Hadia on request. These obligations are set out in the data processing agreement, and every engineer with production access is bound by a confidentiality undertaking.

## 11.3 Security incident notification

| Event                                                                             | Timeline                                                                  |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Codevertex notifies Hadia of a confirmed personal data breach                     | Within 24 hours of confirmation                                           |
| Hadia notifies the Data Protection Commissioner (section 43, Data Protection Act) | Within 72 hours of becoming aware; Codevertex supplies the facts required |
| Affected users notified where there is a real risk of harm                        | Without undue delay, with Hadia's approval of the message                 |
| Written incident report: cause, impact, remediation                               | Within 5 working days                                                     |

## 11.4 Penetration testing

An internal security review against the OWASP Application Security Verification Standard is included in week 11. An independent penetration test by an accredited third party is recommended before launch and annually thereafter. It is commissioned at Hadia's cost at the tester's rate; Codevertex supports the test and remediates all high and critical findings at no charge before launch.

## 11.5 Data processors

| Processor                 | Purpose                                 | Data received                        | Location                         |
|---------------------------|-----------------------------------------|--------------------------------------|----------------------------------|
| Codevertex Africa Limited | Software, hosting operation, support    | All platform data                    | Kenya (operations); EU (servers) |
| Contabo GmbH              | Server infrastructure                   | Platform data stored on its servers  | Germany (EU)                     |
| Cloudflare, Inc.          | Edge network, firewall, offsite backups | Traffic in transit; backup files     | Global edge; EU and US           |
| PayHero                   | Payment collection, wallet, payouts     | Payer number, amount, payout details | Kenya                            |
| Safaricom PLC             | M-Pesa                                  | Payer and recipient number, amount   | Kenya                            |
| Africa's Talking          | SMS delivery                            | Recipient number, message text       | Kenya                            |
| Meta Platforms            | WhatsApp messaging                      | Recipient number, message text       | Global                           |

## SECTION 12

# Hosting, backup and disaster recovery

This section states where Hadia's data is held, how it is backed up and how service is restored after a failure.

## 12.1 Hosting location

| Component                        | Provider and location                                                                                                                              | Content                                                                            |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Application and database servers | Contabo Cloud VPS, European Union region (Germany).<br>Dedicated virtual server: 12 vCPU, 47 GB RAM, 484 GB NVMe storage, running Kubernetes (k3s) | hadia-ui, hadia-api, treasury, authentication, notifications; PostgreSQL databases |
| Offsite backups                  | Cloudflare R2 object storage, separate provider and infrastructure; encrypted at rest                                                              | Nightly encrypted database backups                                                 |
| Edge network                     | Cloudflare (TLS termination, CDN, web application firewall)                                                                                        | Traffic in transit and cached public assets; no database content                   |
| Email                            | Codevertex mail server (mx1.codevertexafrica.com) on the same platform                                                                             | Outbound email; no third-party email processor                                     |
| Funds                            | PayHero, Kenya                                                                                                                                     | Gift funds and payment records. Funds never reside on Codevertex infrastructure.   |

Hosting outside Kenya is a cross-border transfer under sections 48 to 50 of the Data Protection Act. The transfer is documented in the DPIA with its safeguards: provider security certification, encryption in transit and at rest for backups, access restriction and a data processing agreement. Hosting in a Kenyan data centre is available on request and would be quoted as a change to the hosting cost.

## 12.2 Backup schedule

| Backup                   | Frequency        | Scope                                                         | Storage                                                                                                 | Retention                                                                               |
|--------------------------|------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Platform recovery backup | Daily, 05:00 EAT | All platform databases, verified for integrity after each run | Server disk and Cloudflare R2 (offsite)                                                                 | Each new copy replaces the previous one only after it is verified and confirmed offsite |
| Hadia tenant backup      | Daily, 05:00 EAT | Hadia's registry and treasury data only, encrypted            | Cloudflare R2 (offsite), with optional mirror to Hadia's own storage (Google Drive, OneDrive, S3, SFTP) | 7 daily and 4 weekly copies                                                             |
| Payment records          | Continuous       | Every collection and payout                                   | PayHero and Safaricom, independently of Codevertex                                                      | Provider retention                                                                      |

## 12.3 Recovery objectives

| Objective                      | Target     | Meaning for Hadia                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recovery point objective (RPO) | 24 hours   | In the worst case, total loss of the server, changes made since the last nightly backup are restored from other sources. Contributions and payouts in that window are recovered from PayHero's transaction records and re-reconciled, so no payment record is lost. Registry edits, item changes and reservations made in that window may need to be re-entered. |
| Recovery time objective (RTO)  | 4 hours    | Service is restored on a replacement server within 4 hours of a declared disaster.                                                                                                                                                                                                                                                                               |
| Optional enhanced RPO          | 15 minutes | Continuous database log archiving to offsite storage, quoted on request.                                                                                                                                                                                                                                                                                         |

## 12.4 Restore testing

A restore drill is performed quarterly: the latest offsite backup is restored into an isolated environment, record counts and ledger balances are verified against production, and a short report is shared with Hadia. The first drill takes place before launch, in week 11.

## 12.5 Platform outage response

| Scenario                          | Impact                                  | Response                                                                                                                                                                                 |
|-----------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Service or pod failure            | Seconds to minutes                      | Automatic restart; two replicas per service; health checks every 2 minutes with alerting                                                                                                 |
| Server failure or provider outage | Platform unavailable                    | Automated rebuild on a replacement server from version-controlled configuration and backed-up encryption keys, followed by restore of the latest backup and reconciliation with PayHero. |
| During any outage                 | Guests cannot contribute; payouts pause | Funds remain safe in Hadia's PayHero wallet. Pending payouts resume automatically on recovery. Hadia is notified within 30 minutes of a confirmed incident and updated every 2 hours.    |

Capacity scales by adding nodes to the cluster under a documented procedure, without service interruption. Additional capacity is planned with Hadia as monthly contributions grow.

## SECTION 13

# Legal and regulatory compliance

This chapter sets out the legal obligations applicable to Hadia and the controls that address them. It does not constitute legal advice. Items marked **Confirm** require an opinion from Hadia's advocate before launch.

## 13.1 Roles and responsibilities

| Party                                              | Role                                                                                                     | Obligations                                                                                                                          |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Hadia                                              | Service operator and data controller; holds customer funds for beneficiaries through its PayHero account | User terms, fees, payouts, complaints, ODPC registration, tax, KYC and AML cooperation                                               |
| Codevertex Africa Limited                          | Software provider and data processor                                                                     | Secure development and operation, processing on Hadia's instructions under a data processing agreement, breach notification to Hadia |
| PayHero                                            | Payment service provider                                                                                 | Collections, wallet custody, payouts, KYC and AML, card processing                                                                   |
| Africa's Talking, Meta (WhatsApp), SMTP email host | Sub-processors                                                                                           | Message delivery under their terms                                                                                                   |

## 13.2 Applicable law

| Law                                                                                | Obligation                                                                                                                                                                                        | Control                                                                                                                                                                             |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constitution of Kenya, Article 31; Data Protection Act, 2019 and Regulations, 2021 | Lawful and minimal processing, ODPC registration, DPIA, data subject rights, 72-hour breach notification, transfer safeguards, consent for direct marketing, parental consent for children's data | Privacy notice, consent records, data minimisation, DPIA, ODPC registration, data request workflow, breach procedure, documented hosting location, opt-in marketing                 |
| Consumer Protection Act, 2012                                                      | No misleading representations; for internet agreements, disclosure before acceptance, express acceptance, error correction and a copy of the agreement                                            | Fees and payout timing disclosed before sign-up and payment, versioned click-through acceptance, downloadable terms, verified marketing claims                                      |
| Competition Act, 2010                                                              | No false or misleading representations, including reviews and testimonials                                                                                                                        | Consented testimonials only; claims supported by records                                                                                                                            |
| National Payment System Act, 2011 and Regulations                                  | Payment services and holding funds for others are regulated by the Central Bank of Kenya                                                                                                          | Funds held and paid by PayHero; capped hold periods and balances; per-registry ledgers; hourly reconciliation. <b>Confirm</b> whether Hadia requires CBK authorisation or guidance. |
| Proceeds of Crime and Anti-Money Laundering Act, 2009                              | Prevent laundering; maintain records; cooperate with reporting institutions                                                                                                                       | KYC tier 3, verified payout numbers, contribution and balance caps, transaction monitoring, record retention, designated compliance contact                                         |

|                                                                                 |                                                                                                    |                                                                                                                                                        |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Computer Misuse and Cybercrimes Act, 2018                                       | System and data protection; handling of unlawful content                                           | Security controls, audit logging, moderation and reporting, incident response                                                                          |
| Kenya Information and Communications Act; Consumer Protection Regulations, 2010 | Validity of electronic records; no unsolicited messages; opt-out                                   | Logged acceptance, transactional messaging by default, STOP and unsubscribe on marketing, registered sender ID                                         |
| Tax Procedures Act, 2015; VAT Act, 2013; eTIMS                                  | Record retention of at least five years; electronic tax invoices; VAT registration above threshold | Ledger in treasury, gifts recorded as liabilities, commission and premium as income, eTIMS invoicing. <b>Confirm</b> VAT and excise treatment of fees. |
| Children Act, 2022; Data Protection Act, section 33                             | Parental consent and best interests for children's data                                            | Owners aged 18 or over, parental confirmation before child names or photos, unlisted registries by default                                             |
| Persons with Disabilities Act; Constitution, Article 54                         | Accessible public services                                                                         | WCAG 2.2 AA                                                                                                                                            |
| Copyright Act, 2001; Trade Marks Act                                            | Respect for third-party content and marks                                                          | Licensed fonts and images, credited product thumbnails with takedown, descriptive use of store names only                                              |
| Business Registration Service Act; Companies Act, 2015                          | Trade under registered name with business identification                                           | Registered name, number, address and contacts on every page and receipt                                                                                |
| EU and UK GDPR; US CAN-SPAM and COPPA (diaspora users)                          | Equivalent privacy rights, email unsubscribe, no data from children under 13                       | Kenyan controls applied, prior consent for optional cookies, unsubscribe in every email, service not directed at children                              |

### 13.3 Compliance checklist

| # | Requirement       | Implementation                                                                                                                          | Owner                               |
|---|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1 | Privacy policy    | Controller, processors, purposes, lawful bases, retention, rights, transfers and contacts. Linked from every page, form and message.    | Codevertex draft; advocate approval |
| 2 | Terms of service  | Owner, guest and escrow terms; acceptable use; fees; liability; disputes; Kenyan law. Versioned acceptance; 14 days' notice of changes. | Codevertex draft; advocate approval |
| 3 | Refund policy     | Refund grounds, timelines, premium refunds within 14 days if unused, fee treatment. Enforced by the refund workflow.                    | Hadia                               |
| 4 | Cookie policy     | All cookies listed by category, purpose and duration. Strictly necessary cookies only at launch.                                        | Codevertex                          |
| 5 | Cookie consent    | No pre-selected options; equal reject and accept controls; settings available from the footer.                                          | Codevertex                          |
| 6 | Form consents     | Purpose stated for each field; separate unticked consents; consent versioned and withdrawable.                                          | Codevertex                          |
| 7 | Data minimisation | Guests provide name and number only. No identity numbers, dates of birth or location tracking.                                          | Codevertex                          |

|    |                           |                                                                                                                                               |            |
|----|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 8  | Third-party audit         | No advertising trackers. Register of third parties and data shared. Analytics only with consent.                                              | Codevertex |
| 9  | No dark patterns          | No pre-ticked boxes, false urgency or confirm-shaming. Cancellation and deletion as simple as sign-up.                                        | Codevertex |
| 10 | No hidden fees            | Guests see the processing fee and total before paying. Owners see commission and payout terms before creating a registry and on every payout. | Codevertex |
| 11 | Genuine reviews           | Testimonials published only with written consent, retained on file.                                                                           | Hadia      |
| 12 | Substantiated claims      | Marketing statements reflect actual funds handling and payout terms.                                                                          | Hadia      |
| 13 | Alternative text          | Required for uploaded photos; descriptive text on all content images.                                                                         | Codevertex |
| 14 | Colour contrast           | 4.5:1 for text and 3:1 for controls.                                                                                                          | Codevertex |
| 15 | Keyboard navigation       | Full keyboard operation, visible focus, managed dialogs, skip link, automated accessibility testing.                                          | Codevertex |
| 16 | Business details          | Registered name, number, address and contacts in the footer and on receipts.                                                                  | Codevertex |
| 17 | Children's data consent   | Owner age confirmation and parental consent before child information.                                                                         | Codevertex |
| 18 | Unsubscribe               | One-click unsubscribe in email; STOP in SMS and WhatsApp marketing.                                                                           | Codevertex |
| 19 | Licensed fonts and images | Open-licence fonts and licensed imagery, recorded in a licence register.                                                                      | Codevertex |
| 20 | Data deletion             | Self-service export and deletion with deadline tracking and statutory retention.                                                              | Codevertex |

### 13.4 Website content review

| Current text                                   | Issue                                                  | Recommendation                                                                             |
|------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------|
| "Loved by 400+ families"                       | Unverified figure                                      | Remove until supported by platform data                                                    |
| Achieng and David testimonial                  | Not a verified customer statement                      | Replace with consented pilot testimonials                                                  |
| "Funds go straight to you"                     | Funds are held and paid out by rule, net of commission | "Cash gifts are held securely and paid to your M-Pesa automatically, less Hadia's 4% fee." |
| "Your money, kept safe"                        | Requires substantiation                                | Link to a page describing custody, ledgers and reconciliation                              |
| "You only ever share in what's actually given" | Incomplete fee disclosure                              | "Free to create. 4% of cash gifts. Optional premium at KES 300 per event."                 |
| "Carrefour, Naivas, TikTok Shop"               | Third-party marks                                      | Text references only; no logos or implied partnership                                      |

## 13.5 Record retention

| Record                                            | Retention                                                         | Basis                                  |
|---------------------------------------------------|-------------------------------------------------------------------|----------------------------------------|
| Contributions, payouts, refunds, invoices, ledger | 7 years                                                           | Tax Procedures Act; AML record keeping |
| Owner accounts and registries                     | Until deletion, or 24 months after last activity, then anonymised | Data minimisation                      |
| Reservation contact details                       | 12 months after the event                                         | Reservation and thank-you management   |
| One-time passwords                                | 5 minutes, hashed                                                 | Security                               |
| Consent and terms acceptance                      | Account life plus 5 years                                         | Evidence of consent                    |
| Security and audit logs                           | 12 months                                                         | Incident investigation                 |

## 13.6 Legal deliverables

- Privacy notice, cookie policy, terms of service, refund policy, acceptable use policy
- Data processing agreement and sub-processor list
- Data protection impact assessment and record of processing activities
- Breach response procedure and data request procedure
- Third-party and licence registers, accessibility statement

## SECTION 14

# Project delivery

Delivery follows Agile Scrum with weekly demonstrations on the staging environment.

## 14.1 Methodology

| Practice           | Approach                                                                                               |
|--------------------|--------------------------------------------------------------------------------------------------------|
| Iterations         | One-week Sprint 0, four two-week sprints, one-week Sprint 5, hardening week, launch week               |
| Demonstrations     | Weekly 30-minute review with the product owner; decisions logged in the backlog                        |
| Backlog            | User stories traced to FR and NFR identifiers, with acceptance criteria                                |
| Definition of done | Code reviewed, tests and accessibility checks passed, documented, deployed to staging, accepted        |
| Environments       | Development, staging and production on the Codevertex platform                                         |
| Quality assurance  | Unit, integration, end-to-end and contract tests; load testing before launch                           |
| Release management | CI on GitHub Actions; GitOps deployment through ArgoCD                                                 |
| Reporting          | Weekly written status: progress, next steps, risks, decisions required                                 |
| Change control     | Equal-size scope substitutions within the fixed price; other changes estimated and approved in writing |

## 14.2 Project team

| Role                               | Allocation                                  |
|------------------------------------|---------------------------------------------|
| Project lead and business analyst  | Part time, weeks 1 to 12                    |
| Full-stack engineers (Go, Next.js) | Two, weeks 1 to 12                          |
| Product designer                   | Part time, weeks 1 to 5; reviews thereafter |
| QA engineer                        | Part time, weeks 3 to 12                    |
| Platform and DevOps engineer       | As required                                 |

## 14.3 Implementation schedule

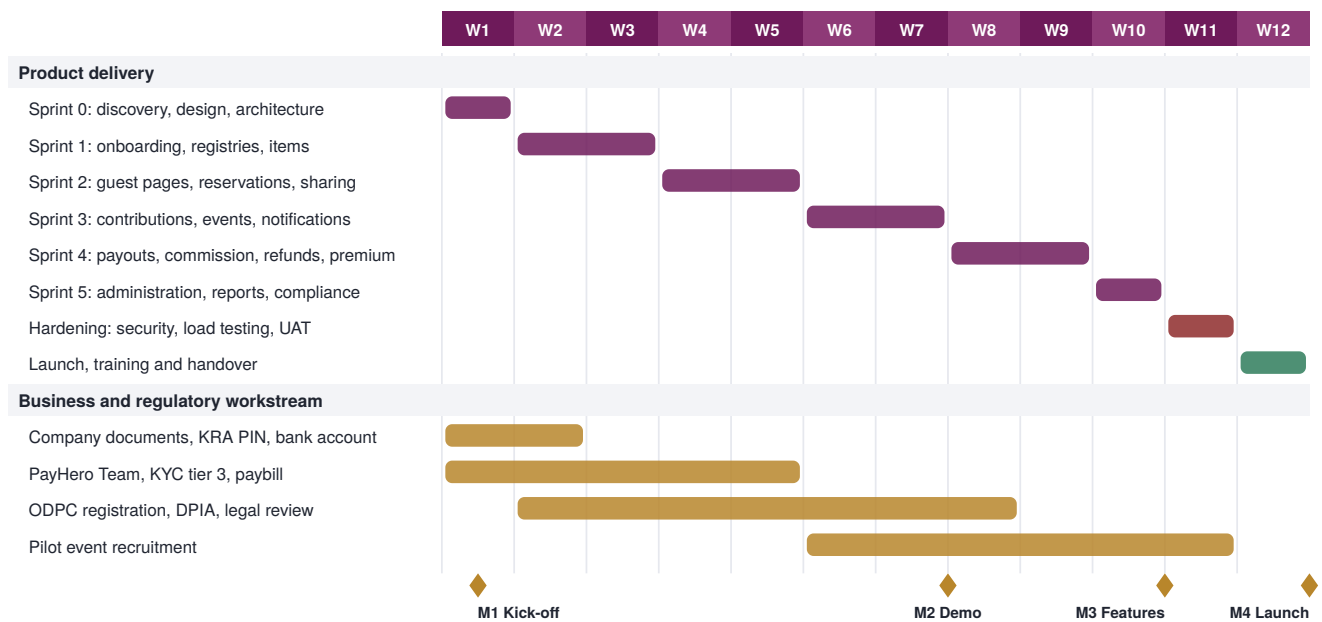


Figure 14. Implementation schedule

| Week   | Phase     | Deliverables                                                                                                                                             |
|--------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Sprint 0  | Kick-off, backlog, UX flows, visual design, architecture approval, environments, legal pack outline                                                      |
| 2 to 3 | Sprint 1  | Mobile sign-in, owner onboarding, registries, manual items, image pipeline, dashboard, tenant configuration                                              |
| 4 to 5 | Sprint 2  | Guest pages, reservations, link completion, sharing and previews, escrow account creation, payout number verification                                    |
| 6 to 7 | Sprint 3  | Contributions, event processing, gift log, receipts and notifications, statements. <b>Milestone 2:</b> end-to-end demonstration with a live contribution |
| 8 to 9 | Sprint 4  | Automated payouts, commission, sweep, contribution limits, approvals, cancellation and refunds, premium                                                  |
| 10     | Sprint 5  | Administration console, exception queue, reports, consent and data requests, accessibility. <b>Milestone 3:</b> feature complete                         |
| 11     | Hardening | Security review, load test, UAT with pilot events, legal pack approval                                                                                   |
| 12     | Launch    | Production release, training, handover. <b>Milestone 4:</b> launch acceptance                                                                            |

Based on contract signature by 16 October 2026, work begins on 19 October 2026 and launch is scheduled for the week of 4 January 2027. Public holidays are accounted for.

## SECTION 15

# Testing and acceptance

| Test               | Objective                                                   | Exit criteria                                                           |
|--------------------|-------------------------------------------------------------|-------------------------------------------------------------------------|
| Automated tests    | Business rules and payment logic                            | All tests pass on every merge                                           |
| End-to-end tests   | Owner, guest, refund and premium journeys                   | Pass on Android Chrome, iOS Safari and desktop                          |
| Live payment tests | Contribution, payout, refund and sweep                      | KES 1 transactions reconcile with PayHero and the ledger                |
| Load test          | Peak traffic                                                | 2,000 contributions per hour and 50 page views per second within NFR-02 |
| Security review    | OWASP risks, authentication, payout changes, link retrieval | No open high or critical findings                                       |
| Accessibility      | WCAG 2.2 AA                                                 | Automated and manual checks passed                                      |
| User acceptance    | Business acceptance by Hadia and pilot users                | All M requirements accepted; no open severity 1 or 2 defects            |

## SECTION 16

# Commercial terms

## 16.1 Fixed price

| Work package                                                                                            | KES            |
|---------------------------------------------------------------------------------------------------------|----------------|
| Discovery, UX and visual design, architecture                                                           | 20,000         |
| Registry service: registries, items, reservations, link completion, gift log, statistics                | 55,000         |
| Web application: website, guest pages, owner dashboard, administration console                          | 55,000         |
| Funds integration: escrow, contributions, automated payouts, commission sweep, limits, refunds, premium | 45,000         |
| Mobile sign-in, message templates and preferences                                                       | 15,000         |
| Compliance features, legal pack drafts, accessibility                                                   | 20,000         |
| Testing, security review, UAT, launch and handover                                                      | 30,000         |
| <b>Total (inclusive of taxes)</b>                                                                       | <b>240,000</b> |

## 16.2 Payment schedule

| Milestone                   | Due            | Share       | KES            |
|-----------------------------|----------------|-------------|----------------|
| M1 Contract signature       | Week 0         | 30%         | 72,000         |
| M2 End-to-end demonstration | End of week 7  | 30%         | 72,000         |
| M3 Feature complete         | End of week 10 | 30%         | 72,000         |
| M4 Launch acceptance        | End of week 12 | 10%         | 24,000         |
| <b>Total</b>                |                | <b>100%</b> | <b>240,000</b> |

## 16.3 Monthly service fee

| Service                                                                                             | KES per month |
|-----------------------------------------------------------------------------------------------------|---------------|
| Treasury platform subscription: ledger, escrow, payouts, approvals, reconciliation, eTIMS invoicing | Included      |
| Platform services: authentication, notifications engine, events and job scheduling                  | Included      |
| Monitoring, security updates and incident response                                                  | Included      |
| Dedicated support and maintenance: support escalation, defect correction, configuration changes     | Included      |
| <b>Total monthly service fee (inclusive of taxes)</b>                                               | <b>8,000</b>  |

The monthly service fee is billed from the launch month.

| Term         | Position                                                                                                                                    |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Usage limits | No per-transaction charge from Codevertex. The fee covers up to 20,000 contributions and 10,000 active registries per month under fair use. |

|               |                                                                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volume growth | Above the fair-use level, server capacity is reviewed with Hadia. Any additional capacity is charged as hosting, at cost, with 60 days' written notice and Hadia's agreement. |
| Price changes | Fixed for the first 12 months from launch. Thereafter it may be adjusted once a year by no more than 10%, with 60 days' written notice.                                       |

## 16.4 Client-borne costs

The following costs are separate from the build price and the monthly service fee and are paid by Hadia.

| Item             | Scope                                                                                                            | Rate (KES)             |
|------------------|------------------------------------------------------------------------------------------------------------------|------------------------|
| Hosting          | Application hosting, database, object storage, daily backups and CDN on the Codevertex platform, billed annually | 12,000 per year        |
| Domain           | Registration and annual renewal of the Hadia domain                                                              | Registrar rate         |
| SMS              | One-time passwords, receipts, alerts and reminders; prepaid credit, minimum top-up KES 500                       | 1.00 per SMS           |
| WhatsApp         | Receipts, alerts, thank-you messages and fallback one-time passwords; monthly plan                               | 500 to 2,000 per month |
| Payment provider | PayHero payout fees, and processing fees if Hadia absorbs them                                                   | PayHero rates          |

## 16.5 Payment provider charges

| Charge                              | Rate                                                                                          | Paid by                    |
|-------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------|
| Gift contribution (into the wallet) | 2.9% of the amount                                                                            | Guest by default, or Hadia |
| Payout to an owner or to Hadia      | PayHero payout band; KES 1.16 on a KES 20 payout                                              | Hadia, from the wallet     |
| Premium purchase                    | The cheaper of the paybill band fee (from the service wallet) and the wallet route; see below | Guest by default           |
| Card payments                       | PayHero card rate, confirmed at onboarding                                                    | As for gifts               |

| Payment (KES) | Paybill or till route | Wallet route | Cheaper         |
|---------------|-----------------------|--------------|-----------------|
| 50            | 6.00                  | 2.61         | Wallet          |
| 200           | 6.00                  | 6.96         | Paybill or till |
| 1,000         | 15.00                 | 30.16        | Paybill or till |
| 10,000        | 50.00                 | 291.16       | Paybill or till |

Rates measured on live PayHero transactions, October 2026. Treasury prices both routes for every payment and uses the cheaper one that can run. Gifts always use the wallet route because escrow payouts are made from the wallet. Every fee is recorded on its payment, so Hadia's reports show real costs. Hadia tops up the service wallet from the treasury console.

## 16.6 Complete fee structure

| Fee                 | Payer                    | Payee                    | Amount                          |
|---------------------|--------------------------|--------------------------|---------------------------------|
| Gift contribution   | Guest                    | Registry (escrow)        | Gift amount                     |
| Processing fee      | Guest (default) or Hadia | PayHero                  | 2.9% of the gift                |
| Commission          | Registry owner           | Hadia                    | 4% of gross (configurable)      |
| Premium upgrade     | Registry owner           | Hadia                    | KES 300 per registry            |
| Build price         | Hadia                    | Codevertex               | KES 240,000 one-time            |
| Monthly service fee | Hadia                    | Codevertex               | KES 8,000 per month             |
| Hosting             | Hadia                    | Codevertex               | KES 12,000 per year             |
| Domain              | Hadia                    | Registrar                | Registrar rate per year         |
| SMS and WhatsApp    | Hadia                    | Codevertex notifications | KES 1.00 per SMS; WhatsApp plan |
| Payout fees         | Hadia                    | PayHero                  | PayHero payout band             |

## 16.7 Operating cost examples

| Monthly item (KES)                                     | 1,000 gifts   | 5,000 gifts    | 10,000 gifts   |
|--------------------------------------------------------|---------------|----------------|----------------|
| Monthly service fee                                    | 8,000         | 8,000          | 8,000          |
| Hosting (KES 12,000 per year)                          | 1,000         | 1,000          | 1,000          |
| SMS receipts (1 per gift)                              | 1,000         | 5,000          | 10,000         |
| WhatsApp Pro                                           | 2,000         | 2,000          | 2,000          |
| Payout fees (1 payout per 10 gifts, about KES 25 each) | 2,500         | 12,500         | 25,000         |
| <b>Operating cost, guests pay the processing fee</b>   | <b>14,500</b> | <b>28,500</b>  | <b>46,000</b>  |
| Commission at 4% (average gift KES 2,000)              | 80,000        | 400,000        | 800,000        |
| <b>Commission after costs</b>                          | <b>65,500</b> | <b>371,500</b> | <b>754,000</b> |
| Commission after costs, if Hadia absorbs the 2.9% fee  | 7,500         | 81,500         | 174,000        |

Average gift KES 2,000. The payout fee per payout will be confirmed with PayHero for Hadia's account. Premium revenue is excluded.

## 16.8 Messaging rates

| Plan or item      | Allowance                       | KES             |
|-------------------|---------------------------------|-----------------|
| SMS               | Per SMS of up to 160 characters | 1.00            |
| WhatsApp Basic    | 100 messages per month          | 500 per month   |
| WhatsApp Standard | 500 messages per month          | 1,000 per month |
| WhatsApp Pro      | Unlimited messages              | 2,000 per month |

Guest receipts are sent by SMS, one per contribution. Owner alerts, digests and thank-you messages are sent by WhatsApp. The WhatsApp Pro plan is recommended from launch because it carries no message cap, so WhatsApp cost stays fixed as volume grows.

## 16.9 Optional services

| Service        | Scope                                                                                                   | Billing                           |
|----------------|---------------------------------------------------------------------------------------------------------|-----------------------------------|
| Managed domain | Registration of the Hadia domain on Hadia's behalf, annual renewal, DNS management and SSL certificates | Annually, at the registrar's rate |

Legal fees and marketing are excluded.

## SECTION 17

# Ownership, licensing and exit

Hadia owns its product, its data and its money. Codevertex retains the shared platform that Hadia uses under licence.

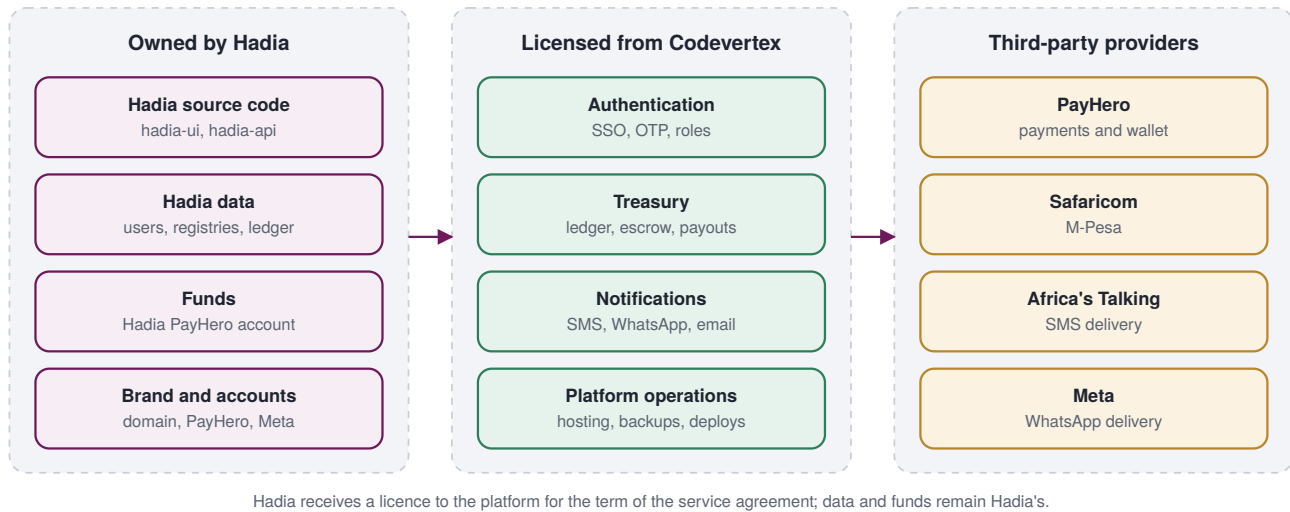


Figure 15. Ownership boundaries

## 17.1 Ownership

| Asset                                                                                                                                       | Owner                       | Terms                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------|
| Hadia source code: hadia-ui, hadia-api, Hadia message templates, designs and content                                                        | Hadia                       | Intellectual property assigned to Hadia on payment of the build price in full      |
| Hadia data: users, registries, items, reservations, contributions, payouts, ledger entries, consents                                        | Hadia                       | Owned by Hadia from creation; Codevertex processes it only on Hadia's instructions |
| Gift funds and commission                                                                                                                   | Hadia (for registry owners) | Held in Hadia's PayHero account, never by Codevertex                               |
| Hadia brand, domain and accounts with PayHero, Safaricom and Meta                                                                           | Hadia                       | Registered in Hadia's name                                                         |
| Codevertex platform: authentication, treasury (ledger, escrow, payouts), notifications, subscriptions, shared libraries, deployment tooling | Codevertex                  | Proprietary; licensed to Hadia as below                                            |

## 17.2 Repository access

The hadia-ui and hadia-api repositories are created in a GitHub organisation owned by Hadia, with Codevertex as a contributor. Hadia has read access from day one and full administrative control at handover. Codevertex platform repositories remain private to Codevertex.

### 17.3 Platform licence

Hadia receives a non-exclusive, non-transferable licence to use the Codevertex platform services for the duration of the service agreement, included in the monthly service fee. Hadia's own developers, or another vendor appointed by Hadia, may use the documented platform APIs with API keys issued to Hadia.

### 17.4 Appointing another developer

Hadia may appoint another developer at any time to maintain or extend hadia-ui and hadia-api. Codevertex provides API documentation, environment access for Hadia's appointed developer and a handover session. The platform services continue to be operated by Codevertex while the service agreement is in force.

### 17.5 Exit and migration

| Step                          | Commitment                                                                                                                                                                    |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Notice                        | Codevertex gives at least 90 days' written notice before discontinuing any platform service Hadia relies on.                                                                  |
| Data export                   | Complete export in open formats: PostgreSQL dump of the Hadia database, ledger and escrow statements (CSV and Excel), user and registry records (JSON), uploaded images       |
| Funds                         | Unaffected. Funds sit in Hadia's PayHero account and continue to be accessible directly through PayHero.                                                                      |
| Transition assistance         | Up to 20 hours of engineering support to move to a replacement payments and ledger service, included; further support at the standard rate                                    |
| Source code escrow (optional) | Platform source code required to run Hadia can be deposited with an independent escrow agent, released to Hadia if Codevertex ceases to trade. Agent fees are borne by Hadia. |

### 17.6 Handover on termination

- Full ownership and administrative control of the hadia-ui and hadia-api repositories
- Final data export and the latest backups, delivered within 10 working days
- Architecture, API and operations documentation, and the legal pack
- Transfer of domain and DNS administration, where managed by Codevertex
- Written confirmation of deletion of Hadia data from Codevertex systems after export, subject to statutory retention

## SECTION 18

# Contract terms

Principal terms of the master services agreement between Hadia and Codevertex. Codevertex's standard terms apply where not varied below.

| Term                            | Position                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initial term                    | 12 months from launch, renewing monthly thereafter                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Termination for convenience     | Either party, after the initial term, with 60 days' written notice                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Termination for cause           | Either party, for a material breach not remedied within 30 days of written notice, or on insolvency                                                                                                                                                                                                                                                                                                                                                                                   |
| Monthly fee after termination   | Payable to the end of the notice period only; no fees thereafter. Export and handover are provided at no charge.                                                                                                                                                                                                                                                                                                                                                                      |
| Service level                   | 99.5% monthly availability for guest pages, checkout and payouts, excluding scheduled maintenance notified 48 hours in advance                                                                                                                                                                                                                                                                                                                                                        |
| Service credits                 | Below 99.5%: 10% of the monthly fee. Below 99.0%: 25%. Below 95.0%: 50%.                                                                                                                                                                                                                                                                                                                                                                                                              |
| Prolonged downtime              | An outage exceeding 72 continuous hours, or service credits in any 2 consecutive months, entitles Hadia to terminate without notice period and receive full handover.                                                                                                                                                                                                                                                                                                                 |
| Delay caused by Codevertex      | 2% of the build price credited for each full week of delay to launch caused solely by Codevertex, up to 10% (KES 24,000). The final milestone is not due until launch acceptance.                                                                                                                                                                                                                                                                                                     |
| Delay caused by Hadia           | Applies where Hadia's decisions, documents, content, feedback, test participation or PayHero and KYC submissions are later than the agreed dates (decisions within 2 working days, documents within 3), or where Hadia changes scope. The schedule extends day for day and no delay credits apply. Milestone payments fall due on their original dates for work completed. Team standby time beyond a cumulative grace of 5 working days is charged at KES 20,000 per week, pro rata. |
| Force majeure and third parties | Events outside both parties' control: natural events, national network or power failures, regulatory action, and processing delays by PayHero, Safaricom or other providers after Hadia has submitted complete documents on time. The schedule extends day for day with no credits or charges to either party. If the event continues beyond 60 days, either party may terminate, and Hadia pays for work completed to that date.                                                     |
| Launch without payments         | Where payment onboarding is the only outstanding item, the platform can launch with reservations; payments are enabled once PayHero is live, and launch acceptance is not withheld on that account.                                                                                                                                                                                                                                                                                   |
| Liability cap                   | Each party's total liability is limited to the fees paid by Hadia in the 12 months preceding the claim. Indirect and consequential losses are excluded.                                                                                                                                                                                                                                                                                                                               |
| Exclusions from the cap         | Breach of confidentiality, the intellectual property indemnity, and wilful misconduct                                                                                                                                                                                                                                                                                                                                                                                                 |
| Codevertex indemnity            | Codevertex indemnifies Hadia against third-party claims that the software infringes intellectual property rights, and against regulatory fines arising from Codevertex's breach of the data processing agreement, within the cap.                                                                                                                                                                                                                                                     |
| Hadia indemnity                 | Hadia indemnifies Codevertex against claims arising from Hadia's content, its terms with users, and its handling of funds.                                                                                                                                                                                                                                                                                                                                                            |

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|               |                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------|
| Governing law | Laws of Kenya. Disputes are referred to negotiation for 30 days, then to mediation, then to the courts of Kenya. |
|---------------|------------------------------------------------------------------------------------------------------------------|

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## SECTION 19

# Risks and mitigations

| Risk                                             | Impact                          | Mitigation                                                                                                    | Owner             |
|--------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------|
| PayHero onboarding exceeds five weeks            | Delayed live payments           | Documents requested in week 1; reservation-only launch available; delay terms allocate time and cost by cause | Hadia, Codevertex |
| Regulatory requirement for payment authorisation | Delay or model change           | Legal opinion in weeks 2 to 6; capped holds and balances; instant payout option                               | Hadia             |
| Fraudulent registries                            | Financial and reputational loss | Verified payout numbers, limits, first-payout hold, monitoring, moderation                                    | Hadia operations  |
| M-Pesa payout failures                           | Owner dissatisfaction           | Automatic reversal and retry, status enquiry, owner notification, exception queue                             | Codevertex        |
| Inconsistent retailer data                       | Slower item entry               | Manual entry as primary method                                                                                | Codevertex        |
| Scope change                                     | Schedule slippage               | Change control; agreed exclusions                                                                             | Both              |
| Holiday period availability                      | Slower decisions                | Holidays scheduled; weekly decision reviews                                                                   | Both              |
| Low cash gifting adoption                        | Revenue below expectation       | Pilots with planners and church networks; conversion analytics                                                | Hadia             |

## SECTION 20

# Post-launch support

| Service            | Terms                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hypercare          | 30 days after launch; same-day response to payment incidents                                                                                                                |
| Warranty           | Defects against this specification corrected at no cost for 90 days                                                                                                         |
| Ongoing support    | Dedicated support and maintenance included in the KES 8,000 monthly service fee                                                                                             |
| Monitoring         | Alerts on payout failures, reconciliation variances, error rates and latency                                                                                                |
| Hosting and domain | Hosting at KES 12,000 per year, billed annually. Domain purchased and renewed by Hadia, or by Codevertex under the optional managed domain service at the registrar's rate. |
| Handover           | Administration guide, finance guide, runbooks, legal pack, registers, architecture documentation                                                                            |
| Enhancements       | Native applications, Kiswahili and referral programmes quoted on request                                                                                                    |

## SECTION 21

# Acceptance

Signature below confirms acceptance of the scope, schedule and price in this document. Changes are subject to the change control process in [Project delivery](#).

| Hadia     | Codevertex Africa Limited |
|-----------|---------------------------|
| Name      |                           |
| Title     |                           |
| Signature |                           |
| Date      |                           |

# Appendix: Client question index

Each question raised by Hadia, the short answer, and the section that sets out the detail.

| Question                                  | Answer                                                                          | Detail                       |
|-------------------------------------------|---------------------------------------------------------------------------------|------------------------------|
| <b>A. OWNERSHIP AND CODE</b>              |                                                                                 |                              |
| Who owns the Hadia source code?           | Hadia, on payment of the build price                                            | Ownership                    |
| Who owns the database and data?           | Hadia; Codevertex is processor only                                             | Ownership                    |
| Will Hadia receive repository access?     | Yes, in Hadia's own GitHub organisation from day one                            | Repository access            |
| What remains Codevertex IP?               | The shared platform services and tooling                                        | Ownership                    |
| What licence does Hadia receive?          | Non-exclusive platform licence for the agreement term                           | Platform licence             |
| Can Hadia appoint another developer?      | Yes, at any time                                                                | Appointing another developer |
| How does Hadia migrate if support stops?  | 90 days' notice, full export, 20 hours transition support, optional code escrow | Exit and migration           |
| What is handed over on termination?       | Repositories, data, backups, documentation, domain                              | Handover                     |
| <b>B. MONEY</b>                           |                                                                                 |                              |
| Complete fee structure, including PayHero | All fees in one table                                                           | Fee structure                |
| Is commission on gross or net?            | On the full gift; the 2.9% processing fee is paid by the guest by default       | Commission basis             |
| Can Hadia change the commission?          | Yes, at any time, globally or per registry                                      | Commission basis             |
| Can Hadia run 0% promotions?              | Yes, scheduled by date range                                                    | Commission basis             |
| Payment reversed after payout?            | Recovered from the owner's future contributions                                 | Reversals                    |
| Who bears fraud losses?                   | Hadia as operator, unless caused by a Codevertex defect                         | Reversals                    |
| Card chargebacks?                         | 14-day card hold; debited from registry, reserve, then owner                    | Reversals                    |
| <b>C. MONTHLY COSTS</b>                   |                                                                                 |                              |
| What is included in KES 8,000?            | Treasury subscription, platform services, monitoring, dedicated support         | Monthly service fee          |
| Can it increase?                          | Fixed 12 months; then at most 10% a year with 60 days' notice                   | Monthly service fee          |
| Volume growth and usage limits?           | No per-transaction fee; fair use to 20,000 contributions a month                | Monthly service fee          |

|                                                       |                                                                           |                       |
|-------------------------------------------------------|---------------------------------------------------------------------------|-----------------------|
| Cost at 1,000, 5,000 and 10,000 contributions?        | KES 14,500, 28,500 and 46,000 a month                                     | Cost examples         |
| <b>D. SECURITY</b>                                    |                                                                           |                       |
| Independent penetration test?                         | Recommended before launch at Hadia's cost; findings fixed free            | Penetration testing   |
| Who can access production?                            | Named Codevertex engineers only, key-based, logged                        | Production access     |
| Can Codevertex staff see customer data?               | Only for support, incidents or restores; logged                           | Staff access          |
| How is privileged access controlled?                  | GitOps changes, sealed secrets, firewall, network policies, MFA           | Production access     |
| Breach notification time?                             | Within 24 hours of confirmation                                           | Incident notification |
| Where is data hosted?                                 | Contabo, EU (Germany); backups on Cloudflare R2                           | Hosting location      |
| Which processors receive data?                        | Seven, listed with purpose and location                                   | Data processors       |
| <b>E. DISASTER RECOVERY</b>                           |                                                                           |                       |
| What does RPO 24 hours mean?                          | Payment records are always recoverable; at most one day of registry edits | Recovery objectives   |
| Backup frequency and location?                        | Daily; on server and offsite in Cloudflare R2                             | Backup schedule       |
| Restore tests?                                        | Quarterly, first in week 11, reported to Hadia                            | Restore testing       |
| If the platform goes down?                            | Restored within 4 hours; funds safe at PayHero                            | Outage response       |
| <b>F. PAYMENTS</b>                                    |                                                                           |                       |
| M-Pesa or PayHero unavailable?                        | Alternative rails offered; payouts queue and resume                       | Failure scenarios     |
| Payout fails?                                         | Funds return to registry and retry automatically                          | Payout controls       |
| Incorrect M-Pesa number?                              | Prevented by OTP and name check; reversal if needed                       | Payout controls       |
| Freeze or manually trigger a payout?                  | Yes, both                                                                 | Payout controls       |
| Owner disputes a payout?                              | Dashboard dispute, 2 working day response                                 | Payout controls       |
| Money when a registry is cancelled?                   | Refunded to each paying number                                            | Cancellation          |
| <b>G. PRODUCT</b>                                     |                                                                           |                       |
| Purchase confirmation, reserved items, manual release | Purchase link, status filters, owner release                              | Lifecycle rules       |
| Transfer, multiple beneficiaries, date change         | Transfer with OTP; one beneficiary; date changes allowed                  | Lifecycle rules       |

|                                                    |                                                                                                                                                                                                                            |                     |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Expiry and reopening                               | Closes 30 days after event; reopen within 90 days                                                                                                                                                                          | Lifecycle rules     |
| <b>H. INTERNATIONAL</b>                            |                                                                                                                                                                                                                            |                     |
| Cards, countries, currencies, FX, fees             | Visa and Mastercard worldwide; mobile money in Kenya, Uganda, Tanzania, Rwanda, Burundi, Ethiopia, DR Congo, Zambia, Malawi, Mozambique, Botswana, South Africa, Nigeria and Ghana; KES registries; no Hadia fee to guests | Supported countries |
| <b>I. CONTRACT</b>                                 |                                                                                                                                                                                                                            |                     |
| Termination rights, notice, fee after termination  | 60 days after initial term; for cause on 30 days' cure; fee ends with notice                                                                                                                                               | Contract terms      |
| Liability cap and indemnity                        | 12 months' fees; IP and data protection indemnity                                                                                                                                                                          | Contract terms      |
| Missed deadline, PayHero delay, prolonged downtime | Codevertex delay: 2% weekly credit to 10%.<br>Hadia delay: extension plus standby cost.<br>Force majeure: extension, no charges.<br>Downtime: termination right after 72 hours                                             | Contract terms      |